Heavy-Tailed Adaptive Quantum Random Walks for Cryptocurrency Market Microstructure: A Reproducible Falsification Benchmark

Mô hình Quantum Random Walk thích nghi ■uôi n■ng cho vi c■u trúc th■ tr■■ng ti■n mã hoá: m■t khung ki■m ch■ng (falsification) có th■ tái l■p

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Code & data availability / Mã ngu■n & d■ li■u: All source code, calibrated parameters, statistical-test outputs and figures referenced in this paper are produced by the accompanying reproducible pipeline (`scripts/phase2`–`scripts/phase7`). Raw tick archives are publicly available from Binance Public Data (`https://data.binance.vision`). Toàn b■ mã ngu■n, tham s■ hi■u ch■nh, k■t qu■ ki■m ■■nh th■ng kê và hình v■ trong bài ■■u ■■■c sinh t■ ■■ng b■i pipeline tái l■p ■i kèm; d■ li■u thô t■i công khai t■ Binance Public Data.

Abstract

Quantum random walks (QRWs) have repeatedly been proposed as a richer alternative to classical stochastic processes for modelling financial price dynamics, motivated by their ballistic spreading, interference structure, and the appeal of quantum-probability formalism. Yet rigorous, reproducible *empirical* tests of QRWs against well-specified classical baselines on real high-frequency limit-order-book (LOB) data remain scarce. This paper develops an **adaptive discrete-time coined QRW** in which the coin operator is driven, *ex ante*, by observable order-book

imbalance (OBI), signed order flow, and trade intensity, with a calibrated dephasing channel representing microstructure information loss. Recognising that a standard fixed-tick shift cannot reproduce the heavy-tailed return distribution of crypto microstructure, we extend the model into a **Heavy-Tailed Adaptive QRW** whose conditional shift samples Pareto-distributed jump magnitudes calibrated by maximum likelihood and capped at the largest move observed in-sample.

We benchmark seven models — QRW Adaptive, QRW Heavy-Tail, three classical random walks (simple, biased, correlated), GARCH(1,1) and geometric Brownian motion (GBM) — on **31 trading days** of tick data for **BTC/USDT, ETH/USDT and BNB/USDT** (over 113 million BTC events), using a strictly chronological, fixed-origin, *ex-ante* protocol. Models are scored on a seven-metric distributional scorecard (Kolmogorov–Smirnov, Wasserstein, variance-scaling exponent, autocorrelation MSE, excess kurtosis, 99% Value-at-Risk and Conditional VaR), and additionally on Diebold–Mariano predictive-accuracy tests, a moving-block bootstrap rank confidence interval, and within-family AIC/BIC model selection.

Our central findings are deliberately reported as a **falsification benchmark** rather than an advocacy of QRWs. (i) On BTC/USDT — the heaviest-tailed asset — the heavy-tailed extension **repairs the catastrophic tail-index pathology** of the fixed-tick QRW: the estimated Hill tail index falls from ≈1.6×10■ to **1.45**, against an empirical value of 2.47 (the magnitude of this repair is asset-dependent; Section 8.5). (ii) The heavy-tailed QRW achieves the **best Wasserstein distance** and the best mean-path Diebold–Mariano loss among all seven models, and the QRW family attains the **highest directional Bernoulli log-likelihood** (best within-family AIC). (iii) Nevertheless, on the aggregate distributional scorecard the QRW variants **do not beat the classical correlated random walk**: the heavy-tailed QRW now *overshoots* excess kurtosis (≈343 vs 75 empirical) and ranks sixth of seven, while a correlated classical walk best matches both the return distribution (Kolmogorov–Smirnov) and the empirical super-diffusive variance-scaling exponent ($\beta \approx 1.3$). The pattern is consistent across the three assets. We conclude that, at the tick scale studied, the quantum-walk analogy buys genuine improvements in *directional/point* prediction but fails *distributional* calibration, and does not dominate a parsimonious classical baseline. We release the full pipeline as a reproducible falsification benchmark for future QRW-in-finance research.

Tóm t■t (Vietnamese)

Quantum random walk (QRW) nhi■u l■n ■■■c ■■ xu■t nh■ m■t thay th■ phong phú h■n cho các quá trình ng■u nhiên c■ ■i■n trong mô hình hoá ■■ng l■c giá, nh■ tính lan truy■n ballistic, c■u trúc giao thoa và s■c h■p d■n c■a hình th■c xác su■t l■■ng t■. Tuy nhiên, các ki■m ■■nh **th■c nghi■m** nghiêm ng■t, có th■ tái l■p, ■■i ch■ng QRW v■i các baseline c■ ■i■n ■■■c ■■c t■ t■t trên d■ li■u s■ l■nh (LOB) t■n su■t cao th■c t■ v■n còn r■t hi■m. Bài báo xây d■ng m■t **QRW thích nghi r■i r■c có coin** trong ■ó toán t■ coin ■■■c ■i■u khi■n *tr■■c (ex ante)* b■i m■t cân b■ng s■ l■nh (OBI), dòng l■nh có d■u và c■■ng ■■ giao d■ch, kèm m■t kênh dephasing hi■u ch■nh ■■i di■n cho m■t mát thông tin vi c■u trúc. Do b■■c d■ch m■t-tick c■ ■■nh không th■ tái t■o phân ph■i l■i su■t ■uôi n■ng, chúng tôi m■ r■ng thành **QRW thích nghi ■uôi n■ng**, trong ■ó b■■c d■ch có ■i■u ki■n l■y m■u biên ■■ nh■y theo phân ph■i Pareto (hi■u ch■nh b■ng h■p lý c■c ■■i) và b■ ch■n trên b■i b■■c nh■y l■n nh■t quan sát trong m■u.

Chúng tôi so chu■n b■y mô hình — QRW Adaptive, QRW Heavy-Tail, ba random walk c■ ■i■n (simple/biased/correlated), GARCH(1,1) và chuy■n ■■ng Brown hình h■c (GBM) — trên **31 ngày** d■ li■u tick c■a **BTC/USDT, ETH/USDT và BNB/USDT** (h■n 113 tri■u s■ ki■n BTC), theo m■t giao th■c ch■t ch■: th■i gian m■t chi■u, g■c c■ ■■nh, hoàn toàn *ex ante*. Các mô hình ■■■c ch■m theo th■ ■i■m phân ph■i b■y ch■ s■ (Kolmogorov–Smirnov, Wasserstein, s■ m■ scaling ph■■ng sai, MSE t■ t■■ng quan, kurtosis v■■t, VaR 99% và CVaR), cùng ki■m ■■nh Diebold–Mariano, kho■ng tin c■y h■ng b■ng moving-block bootstrap, và l■a ch■n mô hình AIC/BIC **trong cùng h■ likelihood**.

K■t qu■ ■■■c trình bày nh■ m■t **khung ki■m ch■ng (falsification)**, không ph■i ■■ c■ v■ QRW. (i) M■ r■ng ■uôi n■ng **s■a ■■■c l■i tail-index th■m ho■** c■a QRW m■t-tick: ch■ s■ ■uôi Hill gi■m t■ ≈1,6×10■ xu■ng **1,45** (empirical 2,47). (ii) QRW ■uôi n■ng ■■t **Wasserstein t■t nh■t** và Diebold–Mariano ■■■ng-trung-bình t■t nh■t trong b■y mô hình, và h■ QRW ■■t **log-likelihood h■■ng (Bernoulli) cao nh■t** (AIC t■t nh■t trong h■). (iii) Tuy nhiên trên th■ ■i■m phân ph■i t■ng h■p, các bi■n th■ QRW **không v■■t ■■■c** random walk t■■ng quan c■ ■i■n: QRW ■uôi n■ng nay *v■■t ng■■ng* kurtosis (≈343 so v■i 75 empirical) và x■p h■ng 6/7, trong khi CRW t■■ng quan kh■p t■t

nh■t c■ phân ph■i l■i su■t (KS) l■n s■ m■ scaling siêu khu■ch tán (β≈1,3). M■u hình này nh■t quán trên ba tài s■n. K■t lu■n: ■ thang tick ■■■c kh■o sát, phép lo■i suy quantum-walk mang l■i c■i thi■n th■t v■ d■ báo *h■■ng/■i■m* nh■ng th■t b■i ■ hi■u ch■nh *phân ph■i*, và không tr■i h■n baseline c■ ■i■n ti■t ki■m tham s■. Chúng tôi công b■ toàn b■ pipeline nh■ m■t benchmark ki■m ch■ng tái l■p ■■■c cho nghiên c■u QRW-trong-tài-chính.

Keywords / T■ khoá: quantum random walk; market microstructure; limit order book; heavy tails; order-book imbalance; decoherence; Diebold–Mariano test; falsification benchmark; cryptocurrency; high-frequency data.

JEL classification: C52, C58, G14. **PACS:** 03.67.-a, 89.65.Gh.

Executive Summary / Tóm t■t ■i■u hành

Five take-aways (EN). 1. **Tail repair (BTC):** the heavy-tailed shift drops the QRW Hill tail index from ≈1.6×10■ to **1.45** (empirical 2.47) — a six-order-of-magnitude correction of the fixed-tick pathology. 2. **Best point/directional predictor:** the QRW family has the **best within-family directional AIC**, the **smallest short-horizon Wasserstein**, and **beats every classical baseline on the Diebold–Mariano mean-path test** ($p < 0.002$, BH-corrected). 3. **Worst distributional fit (BTC/BNB):** on the seven-metric scorecard the QRW variants rank last on BTC; the **correlated classical walk wins**, reproducing both the return distribution (KS) and the empirical super-diffusion (β ≈ 1.3) that the QRW cannot. 4. **Asset-dependence:** on **ETH** (thin-tailed, diffusive) the QRW Heavy-Tail is the **single best model overall** — where the data are diffusive, the diffusive QRW wins. 5. **Honest verdict:** no quantum model dominates a parsimonious classical baseline on distribution; the contribution is a **reproducible falsification benchmark**, not an endorsement.

N■m ■i■m chính (VI). 1. **S■a ■uôi (BTC):** b■■c d■ch ■uôi n■ng kéo tail-index Hill t■ ≈1,6×10■ v■ **1,45** (empirical 2,47) — s■a sai 6 b■c ■■ l■n. 2. **D■ báo ■i■m/h■■ng t■t nh■t:** h■ QRW có **AIC h■■ng trong h■ t■t nh■t**, **Wasserstein ng■n h■n nh■ nh■t**, và **th■ng m■i baseline c■ ■i■n ■ Diebold–Mariano** ($p < 0{,}002$, hi■u ch■nh BH). 3. **Kh■p phân ph■i kém nh■t (BTC/BNB):** trên th■ ■i■m 7 ch■ s■, QRW x■p cu■i ■ BTC; **CRW t■■ng**

quan th■ng, tái t■o c■ phân ph■i (KS) l■n siêu khu■ch tán (β≈1,3) mà QRW không làm ■■■c. 4. **Ph■ thu■c tài s■n:** trên **ETH** (■uôi m■ng, khu■ch tán) QRW Heavy-Tail **t■t nh■t t■ng th■** — n■i d■ li■u khu■ch tán thì QRW khu■ch tán th■ng. 5. **K■t lu■n trung th■c:** không mô hình l■■ng t■ nào tr■i h■n baseline c■ ■i■n v■ phân ph■i; ■óng góp là **benchmark ki■m ch■ng tái l■p**, không ph■i tán d■■ng.

Table of Contents / M■c l■c

List of Abbreviations and Symbols / Danh m■c t■ vi■t t■t và ký hi■u

Symbol	Meaning (EN)	Ý ngh■a (VI)
■_C, ■_P	Coin and position Hilbert spaces	Không gian Hilbert coin và v■ trí
|↑>, |↓>	Coin basis states (up/down)	Tr■ng thái c■ s■ coin (lên/xu■ng)
C(θ)	Coin operator with mixing angle θ	Toán t■ coin v■i góc tr■n θ
S	Conditional shift operator	Toán t■ d■ch có ■i■u ki■n
U = S(C(x)I)	One-step evolution operator	Toán t■ ti■n hoá m■t b■■c

ρ	Density matrix	Ma tr■n m■t ■■
■_η	Dephasing (decoherence) channel, strength η	Kênh dephasing, c■■ng ■■ η
$P(x,t)$	Position distribution after t steps	Phân ph■i v■ trí sau t b■■c
OBI_t	Order-book imbalance at time t	M■t cân b■ng s■ l■nh t■i t
$\theta_t = \pi/4 + \alpha\cdot OBI_t$	Adaptive coin angle	Góc coin thích nghi
α	Coin sensitivity to features	■■ nh■y coin v■i ■■c tr■ng
β	Variance-scaling exponent (Var ~ t■)	S■ m■ scaling ph■■ng sai
α_{tail}	Hill tail index	Ch■ s■ ■uôi Hill
p_{move}	Unconditional one-step move probability	Xác su■t di chuy■n m■t b■■c
δp	Tick size	B■■c giá (tick)

List of Tables / Danh m■c b■ng

List of Figures / Danh m■c hình

Part I — Introduction / Ph■n I — Gi■i thi■u

1. Introduction / Gi■i thi■u

1.1 Motivation

Financial returns at high frequency exhibit a remarkably stable catalogue of *stylised facts*: heavy-tailed, leptokurtic return distributions; near-zero linear autocorrelation of returns coexisting with long memory in absolute returns and order flow; volatility clustering; and, at short horizons, anomalous (often super-diffusive) variance scaling [Cont 2001; Mandelbrot 1963; Lillo, Mike & Farmer 2005]. Classical models capture these facts only partially and usually one at a time: geometric Brownian motion (GBM) is analytically convenient but Gaussian and thin-tailed; GARCH-type models reproduce volatility clustering and unconditional kurtosis but are built on a Gaussian (or Student-t) innovation and a calendar clock that sits uneasily with event-driven microstructure [Bollerslev 1986; Engle 1982]; correlated random walks introduce short-range directional memory but no genuine heavy tails.

Against this backdrop, **quantum random walks (QRWs)** — the quantum-mechanical analogue of classical random walks — have been proposed as a structurally richer modelling primitive [Aharonov, Davidovich & Zagury 1993; Kempe 2003; Venegas-Andraca 2012]. A coined QRW evolves a complex *amplitude* over coin(x)position space by a unitary operator; probabilities arise only at measurement, via the Born rule. Because amplitudes interfere, a QRW spreads *ballistically* (standard deviation ~ t) rather than *diffusively* (~ √t), and a tunable *decoherence* channel interpolates continuously between the quantum (ballistic) and classical (diffusive) regimes [Kendon 2007]. These properties have made QRWs attractive to the quantum-finance literature [Baaquie 2004; Orús, Mugel & Lizaso 2019; Piotrowski & S■adkowski 2004] and to quantum models of decision and cognition that underpin behavioural-finance analogies [Busemeyer & Bruza 2012; Haven & Khrennikov 2013].

> **Tóm t■t 1.1 (VI):** L■i su■t t■n su■t cao có các "stylised facts" ■n ■■nh: ■uôi n■ng, kurtosis cao, t■ t■■ng quan l■i su■t g■n 0 nh■ng trí nh■ dài ■ |l■i su■t| và dòng l■nh, gom c■m bi■n ■■ng, và scaling ph■■ng sai b■t th■■ng ■ thang ng■n. Mô hình c■ ■i■n ch■ b■t ■■■c t■ng ph■n. QRW — phép lo■i suy l■■ng t■ c■a random walk — h■p d■n vì lan truy■n ballistic (~

t) và có tham s■ decoherence n■i suy gi■a l■■ng t■ và c■ ■i■n. Bài này ki■m ■■nh QRW m■t cách th■c nghi■m, nghiêm ng■t.

1.2 The gap: claims outpace reproducible evidence

Two problems recur in the QRW-in-finance literature. First, the **mapping** from quantum objects to market observables is often left as a metaphor: what, precisely, is the "coin", the "position", the "decoherence rate", in terms of quantities a microstructure researcher can measure from a limit order book? Second, **evaluation** is frequently either purely theoretical or conducted against weak baselines (e.g., a single GBM) on small samples, with no correction for multiple testing, no out-of-sample protocol, and no predictive-accuracy significance test. The result is a literature in which the *appeal* of quantum formalism is not matched by falsifiable, reproducible empirical evidence.

This paper takes the opposite stance, in the spirit of the project's theory notes: *the value of the framework must be judged by its ability to produce testable, scoreable forecasts, not by the attractiveness of the metaphor.* We therefore (a) fix an explicit, defensible mapping with calibration protocols and validity ranges for every parameter; (b) attach a classical baseline to every quantum interpretation so that the benefit of coherence can be *falsified*; and (c) evaluate on a large, multi-asset, multi-day sample with corrected statistical tests.

1.3 Contributions

An adaptive, ex-ante QRW for the LOB. We specify a discrete-time coined QRW whose coin angle is an affine-then-clamped function of standardized, *causally* computed microstructure features (order-book imbalance, signed tick direction, OBI change, |OBI|, log trade intensity), with a density-matrix dephasing channel. All features used to forecast a holdout path are computed only from information available before the forecast origin.

A heavy-tailed shift that repairs the tail pathology. We introduce a Heavy-Tailed Adaptive QRW whose conditional shift draws Pareto-distributed jump magnitudes (tail index fitted by maximum likelihood; minimum jump = tick size; magnitude capped at the in-sample maximum move), integrated as a distinct seventh model in the benchmark.

A rigorous, reproducible, multi-asset benchmark. Thirty-one trading days of BTC/ETH/BNB tick data; a fixed-origin, chronological, ex-ante protocol; a seven-metric distributional scorecard; Diebold–Mariano tests with Newey–West HAC variance and Benjamini–Hochberg false-discovery control; moving-block bootstrap rank intervals; and within-likelihood-family AIC/BIC selection that corrects a common cross-family comparison error.

A falsification result, reported honestly. The heavy-tailed QRW fixes the tail index and wins on Wasserstein/point-prediction and directional likelihood, but overshoots kurtosis and does not beat a correlated classical walk on the aggregate distributional scorecard, consistently across assets. We frame and release the artefact as a reproducible falsification benchmark.

> **Tóm t■t 1.3 (VI):** B■n ■óng góp: (1) QRW thích nghi ex-ante cho LOB v■i coin ■i■u khi■n b■ng ■■c tr■ng vi c■u trúc tính nhân qu■; (2) b■■c d■ch ■uôi n■ng Pareto s■a l■i tail-index, thêm vào benchmark nh■ mô hình th■ 7; (3) benchmark ■a tài s■n, ■a ngày, nghiêm ng■t v■i DM test (HAC + Benjamini–Hochberg), bootstrap h■ng, và AIC/BIC trong cùng h■; (4) m■t k■t qu■ falsification trung th■c: QRW ■uôi n■ng s■a tail và th■ng ■ d■ báo ■i■m nh■ng không v■■t CRW t■■ng quan ■ phân ph■i.

1.4 Research questions and falsifiable hypotheses

Following the project's "rejectable-hypotheses" discipline, we state in advance what would *falsify* a claim of QRW usefulness. Each hypothesis pairs the quantum model with a classical baseline.

#	Hypothesis (alternative)	Falsified if …	Verdict (this study)
H1	An OBI-adaptive coin improves *directional* prediction over a fixed coin / classical walk	QRW within-family directional AIC not better than CRW	**Not falsified** — QRW has best directional AIC (Table 7)

H2	The QRW reproduces empirical *super-diffusive* variance scaling (β > 1)	Calibrated QRW β ≈ 1 (diffusive)	**Falsified** — QRW β ≈ 1.0; super-diffusion matched only by correlated CRW (Table 2)
H3	A heavy-tailed shift yields a *finite, plausible* tail index	Simulated tail index remains divergent or wildly off	**Not falsified on BTC** (1.45 vs 2.47); muted on ETH/BNB (Section 8.5)
H4	The heavy-tailed QRW improves the *aggregate distributional* scorecard over classical walks	QRW does not beat the best CRW on mean scorecard rank	**Falsified** — best CRW outranks both QRWs on BTC/BNB (Tables 5, 10)
H5	The QRW improves *point/path* forecast accuracy (Diebold–Mariano)	QRW not significantly better than classical baselines on mean-path loss	**Not falsified** — QRW Heavy-Tail beats all baselines (Table 12)

The mixed verdict (H1, H3, H5 survive; H2, H4 are falsified) is the substance of the paper and the reason we frame the contribution as a benchmark rather than an endorsement.

> **Tóm t■t 1.4 (VI):** N■m gi■ thuy■t có th■ bác b■, m■i cái ghép QRW v■i baseline c■ ■i■n. K■t qu■: **H1 (coin c■i thi■n d■ báo h■■ng), H3 (■uôi n■ng cho tail-index h■u h■n — rõ trên BTC), H5 (c■i thi■n d■ báo ■i■m/■■■ng)** không b■ bác; **H2 (tái t■o siêu khu■ch tán) và H4 (th■ng th■ ■i■m phân ph■i t■ng h■p)** b■ bác. K■t qu■ h■n h■p này là n■i dung chính và là lý do trình bày nh■ benchmark, không ph■i tán d■■ng.

1.5 Paper organization

Section 2 reviews the relevant literature across microstructure, quantum finance, and heavy-tailed statistics. Section 3 fixes the QRW formalism. Section 4 develops the market mapping and the heavy-tailed extension. Section 5 describes the data pipeline; Section 6 the models; Section 7 the statistical methodology. Section 8 presents results (single-asset BTC and cross-asset). Section 9 discusses insights, Section 10 limitations, and Section 11 concludes. Appendix A is a complete artefact catalogue (the meaning of every result file, figure and script); Appendix B collects extended tables; Appendix C documents reproducibility.

Part II — Literature Review / Ph■n II — T■ng quan tài li■u

2. Literature Review (Background and Related Work) / T■ng quan tài li■u

This section synthesises three strands of literature. Citations use the numbered reference list in Section 12; the 22 core references were independently verified with DOIs.

2.1 Market microstructure and the limit order book

The modern limit order book (LOB) is a double-sided queueing system whose dynamics generate price formation at the finest time scale. Empirical and modelling work has established that order-book state — especially **order-book imbalance (OBI)** — carries short-horizon predictive content for the sign and size of the next price move [Cont, Stoikov & Talreja 2010; Gould et al. 2013]. Statistical regularities of order books, including power-law features in volumes and the long memory of signed order flow, were documented by Bouchaud, Mézard & Potters [2002] and Lillo, Mike & Farmer [2005]. Structural models range from the queue-reactive model of Huang, Lehalle & Rosenbaum [2015] to dynamic equilibrium models of the LOB [Parlour 1998; Ro■u 2009] and stochastic order-book dynamics [Cont, Stoikov & Talreja 2010]. Two classical information-based microstructure models frame the economics of price formation: the sequential-trade model of Glosten & Milgrom [1985] and the strategic-informed-trader model of Kyle [1985]. Our mapping borrows the central empirical lesson of this literature — *OBI and signed flow predict short-horizon direction* — and encodes it directly in the QRW coin.

A persistent empirical lesson is that **order-book imbalance has genuine but short-lived directional predictive power**: the sign and immediate magnitude of the next mid-price move are partially forecastable from the relative depth/flow on the two sides of the book. Cont, Stoikov & Talreja [2010] formalise order-book dynamics as a stochastic queueing system; Gould et al. [2013] survey the LOB as an object of study; Lillo, Mike & Farmer [2005] and Bouchaud, Mézard & Potters [2002] document the long memory of signed order flow and the power-law structure of volumes. Equilibrium/strategic models (Parlour [1998]; Ro■u [2009]) and the queue-reactive model (Huang, Lehalle & Rosenbaum [2015]) explain *why* such predictability arises and *why it decays*. The two canonical information-economics models — Glosten &

Milgrom [1985] (sequential trade, bid–ask spread as adverse-selection compensation) and Kyle [1985] (strategic informed trader, linear price impact) — frame the deeper economics: price moves because order flow is informative. Our coin operationalises this single empirical lesson, encoding imbalance and signed flow directly into the directional response.

2.2 Quantum random walks and quantum finance

The discrete-time coined QRW was introduced by Aharonov, Davidovich & Zagury [1993]; Kempe [2003] and Venegas-Andraca [2012] provide standard reviews, and Kendon [2007] reviews decoherence in quantum walks — the mechanism by which a QRW degrades to a classical walk. QRWs are computational primitives (universal for quantum computation), but here we use only their *stochastic-process* structure: ballistic spreading and interference, tunable toward diffusion by decoherence.

In finance, Baaquie [2004] develops a path-integral / Hamiltonian formulation of option pricing ("quantum finance"), recasting the Black–Scholes equation as a Schrödinger-type evolution and opening a programme of Hamiltonian methods for derivatives and interest rates. Piotrowski & S■adkowski [2004] explore quantum-game and thermodynamic analogies for portfolios, in which strategies are operators and payoffs emerge from non-commuting observables. Orús, Mugel & Lizaso [2019] survey quantum *computing* for finance — portfolio optimisation via quantum annealing/QAOA, amplitude-estimation speed-ups for Monte-Carlo pricing, and machine-learning hybrids — emphasising that near-term advantage is most plausible in optimisation and sampling rather than in forecasting. The quantum-probability programme in decision theory [Busemeyer & Bruza 2012; Haven & Khrennikov 2013] argues that order/context effects and violations of classical probability axioms in human judgement are naturally captured by Hilbert-space models; the stochastic-mechanics bridge of Baez & Biamonte [2012] shows formally how classical Markov dynamics and quantum dynamics share an operator-algebraic skeleton, which is exactly why a *decohered* quantum walk degenerates to a classical Markov walk. These works collectively justify treating quantum amplitudes as *latent modelling constructs* rather than physical claims about markets. **Crucially, direct, reproducible empirical tests of QRWs on tick-level LOB data, against well-specified classical baselines and with predictive-accuracy significance**

testing, remain sparse — the gap this paper addresses, and the reason we pair every quantum interpretation with a falsifiable classical baseline.

2.3 Heavy tails, scaling, and adaptive markets

That asset returns are heavy-tailed and non-Gaussian is among the oldest stylised facts [Mandelbrot 1963], who proposed Lévy-stable laws for cotton prices. This was later sharpened by Mantegna & Stanley [1995] into truncated-Lévy scaling for a stock index, and by Gabaix et al. [2003] into the celebrated "inverse-cubic law" linking the power-law tail of returns (exponent ≈ 3) to the power-law distribution of trade volumes and the square-root price impact of large orders. Cont [2001] synthesises the canonical stylised-facts catalogue — heavy tails, volatility clustering, aggregational Gaussianity, the leverage effect, and the slow decay of absolute-return autocorrelation — that any micro-scale model should be measured against; our seven-metric scorecard is a direct operationalisation of a subset of these facts. Tail indices are commonly estimated by the Hill [1975] estimator, which we adopt (with its known sensitivity to the choice of order-statistic threshold). The adaptive-markets hypothesis [Lo 2004] reframes efficiency as an evolving, context-dependent property — a useful lens for an *adaptive* coin whose responsiveness can be regime-dependent, and a reminder that a single model need not dominate across assets or epochs (consistent with our strong asset-dependence). The methodological backbone of our evaluation rests on Diebold & Mariano [1995] for predictive-accuracy comparison, Newey & West [1987] for HAC variance estimation under serial correlation, Politis & Romano [1994] for the stationary/moving-block bootstrap that respects dependence when resampling, and Benjamini & Hochberg [1995] for false-discovery-rate control across the many pairwise and per-metric tests we run.

> **Tóm t■t 2 (VI):** Ba m■ch tài li■u: (1) Vi c■u trúc LOB — OBI và dòng l■nh có d■u d■ báo h■■ng ng■n h■n (Cont–Stoikov–Talreja, Gould và c■ng s■, Glosten–Milgrom, Kyle); (2) QRW & quantum finance — QRW (Aharonov 1993; Kempe 2003), decoherence (Kendon 2007), quantum finance (Baaquie 2004; Orús và c■ng s■ 2019), nh■ng thi■u ki■m ■■nh th■c nghi■m trên d■ li■u LOB th■t; (3) ■uôi n■ng & scaling — Mandelbrot 1963, Gabaix 2003, Cont 2001, ■■c l■■ng Hill 1975, cùng n■n ph■■ng pháp DM (1995), Newey–West (1987), bootstrap (Politis–Romano 1994), FDR

(Benjamini–Hochberg 1995).

Part III — Methodology / Ph■n III — Ph■■ng pháp nghiên c■u

Methodology overview / T■ng quan ph■■ng pháp. Per the IMRAD convention, this Part bundles the full research design: the QRW formalism (§3), the market mapping and heavy-tailed extension (§4), the data and ex-ante evaluation protocol (§5), the seven competing models (§6), and the statistical test battery (§7). / Theo IMRAD, Ph■n III g■m toàn b■ thi■t k■ nghiên c■u: hình th■c QRW (§3), ánh x■ th■ tr■■ng + m■ r■ng ■uôi n■ng (§4), d■ li■u và giao th■c ■ánh giá ex-ante (§5), b■y mô hình (§6), và b■ ki■m ■■nh th■ng kê (§7).

3. Theoretical Framework: Discrete-Time Coined QRW / Khung lý thuy■t

This section fixes the formalism used throughout. We adopt a single convention: the coin state *up* shifts right, *down* shifts left; the coin is applied **before** the conditional shift; and position is measured only after the final evolution step. Although called a "random walk", between measurements a discrete-time QRW (DTQRW) is a *deterministic, linear, unitary* evolution of complex amplitudes — randomness enters only at measurement via the Born rule. This is the origin of interference and of spreading behaviour different from a classical random walk (CRW).

3.1 Hilbert space and state

The coin space is ■_C = ■^{2} = span{|↑>, |↓>}, and the position space on the integer lattice is ■_P = ■^{2}(■) = span{|x> : x in ■}. The total space is the tensor product ■ = ■_C (x) ■_P. A general state at integer time t is

```
|ψ_t■ = Σ_x ( a_x(t) |↑,x■ + b_x(t) |↓,x■ ), a_x(t), b_x(t) ∈ ■,
```

with normalization $\langle\psi_t|\psi_t\rangle = \Sigma_x (|a_x(t)|^2 + |b_x(t)|^2) = 1$. The probability of measuring the walker at position x after t steps is **$P(x,t) = |a_x(t)|^2 + |b_x(t)|^2$**. Amplitudes that reach the same (coin, position) pair add *at the amplitude level* before squaring; because amplitudes carry phase, they can reinforce or cancel — interference.

3.2 Coin operator

The coin is a unitary C : ■_C → ■_C. The project's canonical coin is the Hadamard operator

```
H = (1/√2) [ 1 1 ; 1 −1 ], H†H = I■.
```

A real one-parameter family controlling directional bias is the reflection-rotation coin

```
C(θ) = [ cos θ sin θ ; sin θ −cos θ ], with C(π/4) = H (det = −1),
```

and, where a determinant +1 rotation is required, $R(\theta) = [\cos\theta, -\sin\theta; \sin\theta, \cos\theta]$. The two families must not be mixed within an experiment without stating the convention, because sign and phase alter the interference structure even when one-step probabilities look identical. The coin acts identically at every position via C (x) I_P.

3.3 Conditional shift and one-step evolution

The shift operator is

```
S = |↑■■↑| ⊗ Σ_x |x+1■■x| + |↓■■↓| ⊗ Σ_x |x–1■■x|,
```

so $S|\uparrow,x\rangle = |\uparrow,x+1\rangle$ and $S|\downarrow,x\rangle = |\downarrow,x-1\rangle$. Since S permutes an orthonormal basis, $S^\dagger S = I$. The one-step operator applies the coin first: **U = S (C (x) I_P)**, $|\psi_{t+1}\rangle = U|\psi_t\rangle$, and $U^\dagger U = (C^\dagger C)$ (x) $I_P = I$. This unitarity invariant — norm equal to one up to machine precision — is the single most important implementation check. For the Hadamard coin the amplitude recurrences are

```
a_x(t+1) = ( a_{x–1}(t) + b_{x–1}(t) ) / √2, b_x(t+1) = ( a_{x+1}(t) – b_{x+1}(t)
) / √2.
```

3.4 Initial state, symmetry, and ballistic spreading

With a symmetric initial coin state $|\chi_{sym}\rangle = (|\uparrow\rangle + i|\downarrow\rangle)/\sqrt{2}$ at the origin, the position distribution is symmetric, $P(x,t) = P(-x,t)$. A Fourier transform diagonalises the translation-invariant evolution into 2×2 blocks $U(k) = S(k)H$ with $S(k) = \mathrm{diag}(e^{ik}, e^{-ik})$; the eigen-phases $e^{i\omega(k)}$ have group velocity $v_g = d\omega/dk$ bounded by $|v| \leq 1/\sqrt{2}$ for the Hadamard walk. Stationary-phase analysis yields two ballistic fronts near $x \approx \pm t/\sqrt{2}$ and a standard deviation that grows **linearly** in t — *ballistic* spreading — in contrast to the $\sqrt{t}$ *diffusive* spreading of a CRW. This $\mathrm{Var} \sim t^2$ ($\sigma \sim t$) versus $\mathrm{Var} \sim t$ distinction is the quantitative signature we test via the variance-scaling exponent β (Section 7).

3.5 Density matrix and the decoherence channel

A pure state has density operator $\rho_t = |\psi_t\rangle\langle\psi_t|$, evolving as $\rho_{t+1} = U \rho_t U^\dagger$. A dephasing channel in the coin–position basis is

```
■_η(ρ) = (1 − η) ρ + η Σ_{c,x} Π_{c,x} ρ Π_{c,x}, Π_{c,x} = |c,x■■c,x|, 0 ≤ η ≤ 1,
```

which is completely positive and trace preserving (CPTP). $\eta = 0$ preserves coherence; $\eta = 1$ erases all off-diagonal terms each step, and the diagonal probabilities then evolve as a Markov chain with transition magnitudes $|C_{ij}|^2$ — for the Hadamard coin, the simple symmetric CRW. Crucially, the naïve mixing rule $\psi_{new} = (1-\gamma)\psi + \gamma\cdot\psi_{classical}$ does **not** preserve norm and is not a valid quantum channel; classical mixtures must be formed at the density-matrix level, $\rho' = (1-\eta)\rho_{coherent} + \eta\ \rho_{dephased}$. Different "noise" mechanisms (coin-only dephasing, position measurement, broken links, random coin) yield *different* classical limits and crossover times, so any decoherence parameter must be reported together with its channel definition.

3.6 Verification invariants

A correct implementation preserves or checks: (1) normalization $\Sigma_x P(x,t) = 1$; (2) unitarity ■U†U − I■_F near machine precision; (3) causality $P(x,t) = 0$ for $|x| > t$ and parity $x + t$ even; (4) symmetry $P(x,t) = P(-x,t)$ for symmetric initial coin; (5) spectrum $|\lambda_j(U)| = 1$ on a finite periodic lattice; (6) pure-state norm ■ψ_t■■ = 1 absent decoherence and boundary absorption. These detect wrong shift direction, off-by-one, boundary leakage and wrong tensor order *before* any financial interpretation. The accompanying notebook (`notebooks/01_theory_verification.ipynb`) checks the symbolic Hadamard identity and the Frobenius unitarity norm on a finite cycle; the corresponding artefacts are `reports/checkpoints/theory_verification_results.json` and `figures/prob_evolution.gif`.

> **Tóm t■t 3 (VI):** DTQRW m■t chi■u: tr■ng thái trên coin(x)v■ trí, coin Hadamard H, d■ch có ■i■u ki■n S, ti■n hoá U = S(C(x)I) là unitary (b■o toàn norm). Coin ■■i x■ng cho phân ph■i ■■i x■ng và lan truy■n **ballistic** ($\sigma \sim t$) — khác CRW ($\sigma \sim \sqrt{t}$); ■ây là d■u hi■u ta ki■m b■ng s■ m■ scaling β. Decoherence ■_η là kênh CPTP: η=0 gi■ l■■ng t■, η=1 ■■a v■ CRW c■ ■i■n. Sáu invariant (chu■n hoá, unitary, nhân qu■, ■■i x■ng, ph■, norm) dùng ■■ ki■m th■ tr■■c khi di■n gi■i tài chính.

3.7 QRW versus CRW: an analytical comparison

The decisive structural difference is summarised below; it is what the variance-scaling test (Section 7.2) operationalises empirically.

Classical baseline. For a simple symmetric random walk $X_T = \Sigma_{t \le T} \xi_t$ with $\xi_t = \pm 1$ i.i.d., ■$[X_T] = 0$, $\mathrm{Var}(X_T) = T$, $\sigma_{CRW}(T) = \sqrt{T}$, and the exact law is binomial on the parity class, $\Pr(X_T = x) = 2^{-T} C(T, (T+x)/2)$. After rescaling by $\sqrt{T}$ the CLT gives a Gaussian limit — **diffusive** scaling, width $O(\sqrt{T})$.

Quantum case. For the coherent Hadamard DTQRW, $|\psi_T\rangle = U$■$|\psi$■$\rangle$, Fourier analysis writes the position-x component as an oscillatory integral in phase $kx - T\omega_j(k)$; stationary points satisfy $x/T = d\omega_j/dk$. Because x/T concentrates on a non-degenerate velocity variable, the typical position scales **linearly** in T: $\sigma_{QRW}(T) = O(T)$, $\mathrm{Var}(X_T) = O(T^2)$ — **ballistic** spreading. For the project's symmetric initial coin, Konno's limit theorem gives the exact quadratic coefficient

$$\lim_{T\to\infty} \mathrm{Var}(X_T)/T^2 = 1 - 1/\sqrt{2} \approx 0.2929, \text{ so } \mathrm{Var}(X_T) = (1 - 1/\sqrt{2})\, T^2 + O(T).$$

(The often-quoted $T^2/2$ coefficient is incorrect for the standard Hadamard walk used here; the verification notebook checks the numerical coefficient directly.) At $T = 100$ the coherent QRW variance is $\approx 0.29 \cdot T^2$ while the CRW variance is T, a ratio ≈ 29 — confirming two different scaling regimes, **not** that the QRW forecasts better.

Property	Symmetric CRW	Coherent Hadamard DTQRW
Evolving object	Non-negative probabilities	Complex amplitudes
Path combination	Add probabilities	Add amplitudes, then square
Evolution	Stochastic / Markov	Unitary until measurement
Std. deviation	$\sqrt{T}$	T
Variance	T	$\approx(1 - 1/\sqrt{2})\, T^2$
Limit rescaling	$X_T/\sqrt{T}$	X_T/T
Typical shape	Single-peak Gaussian	Two fronts + interference
Initial-phase dependence	No	Yes
Per-step measurement	Is the dynamics	Destroys coherence → Markov

Decoherence crossover. Under partial dephasing there is a crossover

```
Var(X_T) ~ c·T² for T ■ τ_dec (quantum, ballistic), D(η)·T for T ■ τ_dec
(classical, diffusive),
```

where τ_dec depends on the channel and noise strength, and D(η) need not equal the standard CRW coefficient. **This is precisely the lens for our empirical finding:** if calibration drives decoherence high and scaling back toward $\beta = 1$, that is a valid result indicating the coherence analogy is unnecessary at the scale studied (Section 2.9 of the project theory notes). Real prices cannot spread ballistically without bound — tick size, finite liquidity, mean reversion and trading halts forbid it — so a purely coherent QRW is a *null model*, not a default empirical description. Section 8.1 shows the calibrated QRW indeed sits at $\beta \approx 1.0$.

> **Tóm t■t 3.7 (VI):** CRW: Var = T (khu■ch tán, √T). QRW coherent: Var ≈ (1–1/√2)T² ≈ 0,293·T² (ballistic, ~T); h■ s■ ■úng là 1–1/√2 ch■ không ph■i 1/2; t■ l■ t■i T=100 ≈ 29. Decoherence t■o **crossover**: ballistic (T■τ_dec) → khu■ch tán (T■τ_dec). ■ây chính là l■ng kính cho phát hi■n th■c nghi■m: n■u calibration ■■y β v■ 1, ngh■a là analogy l■■ng t■ **không c■n thi■t ■** thang này — và M■c 8.1 cho th■y QRW hi■u ch■nh n■m ■ β ≈ 1,0.

3.8 Worked example: three steps of the Hadamard walk

To make the interference concrete, start from the symmetric coin state |χ_sym> = (|↑> + i|↓>)/√2 at x = 0 and apply U = S(H(x)I) for a few steps (amplitudes are tracked exactly).

Step 0: a■ = 1/√2, b■ = i/√2 (all other amplitudes zero).

Step 1: applying H then S, the up-component shifts to x = +1 and the down-component to x = –1: a_{+1} = (a■ + b■)/√2 = (1 + i)/2, b_{–1} = (a■ – b■)/√2 = (1 – i)/2. The position probabilities are P(+1) = |a_{+1}|² = 1/2 and P(–1) = |b_{–1}|² = 1/2 — symmetric, as required.

Step 2: amplitudes now populate x in {–2, 0, +2}. The two histories that arrive at x = 0 (down-then-up vs up-then-down) carry *different phases* and partially cancel, producing the characteristic central dip; the outer positions x = ±2 reinforce. Continuing this recurrence to large T produces the twin ballistic fronts near x ≈ ±T/√2 and the oscillatory interior of Section 3.4. The crucial point for finance: **the same**

superposition that creates these fronts is exactly what a per-step measurement (decoherence) destroys, collapsing the walk to the classical binomial. A market in which prices are "observed" every tick is, in this language, strongly decohered — anticipating the empirical $\beta \approx 1$.

3.9 Implementation, invariants and complexity

The coherent walk is implemented on a density matrix ρ (`src/models/qrw_core.py`), enabling exact dephasing via the operator-sum form ■_η(ρ) = (1−η)ρ + η Σ Π_{c,x} ρ Π_{c,x}. A walk over T steps reaches positions $|x| \le T$, so the lattice has $O(T)$ sites and ρ has $O(T^2)$ entries on the coin(x)position space; one step costs $O(T^2)$ for the pure-state recurrence and up to $O(T^2)$ for the diagonal dephasing, giving $O(T^3)$ for a full T-step pure simulation and $O(T^2)$ per step for the diagonal Markov limit. For the *benchmark* simulator we exploit that, after calibration, the relevant object is the one-step directional probability and move probability; the Monte-Carlo path simulator is therefore O(n_paths · n_steps), independent of lattice size, which is what makes the 7-model, multi-asset benchmark tractable. The six verification invariants of Section 3.6 are asserted in `notebooks/01_theory_verification.ipynb` and the Phase-1 checkpoint artefacts; they guard against the classic implementation faults (wrong shift sign, off-by-one parity, boundary leakage, wrong tensor order) that can pass a naïve normalization check yet corrupt the financial interpretation.

> **Tóm t■t 3.8–3.9 (VI):** Ví d■ ba b■■c Hadamard cho th■y hai l■ch s■ t■i cùng v■ trí có **pha khác nhau** nên tri■t tiêu m■t ph■n (t■o "dip" trung tâm) và t■ng c■■ng ■ biên (hai front ±T/√2). Chính ch■ng ch■p này b■ **phép ■o m■i b■■c (decoherence) phá hu■**, ■■a v■ binomial c■ ■i■n — th■ tr■■ng "quan sát m■i tick" là h■ decohered m■nh (d■ báo β≈1). Tri■n khai b■ng density matrix (O(T²) ô, dephasing operator-sum); nh■ng b■ mô ph■ng benchmark ch■ O(n_paths·n_steps) nên ch■y ■■■c ■a mô hình, ■a tài s■n. Sáu invariant ■■■c ki■m trong notebook ■■ ch■ng l■i shift/parity/boundary/tensor.

4. Market Mapping and the Heavy-Tailed Extension / Ánh x■ th■ tr■■ng và m■ r■ng ■uôi n■ng

4.1 Philosophy of the mapping

The mapping is a *quantum-walk-inspired mathematical model*, not a claim that the order book is a physical quantum system. "Amplitude", "coin" and "decoherence" are **latent modelling constructs**. Three principles are enforced: (1) every QRW object maps to an observable or estimable market quantity; (2) every parameter has a calibration protocol and a validity range; (3) every interpretation has a corresponding classical baseline so the benefit of coherence can be *falsified*.

QRW component	Market counterpart	Justification	Caveat
x in ■	Price displacement in ticks vs reference	LOB has a discrete tick grid	Mid-price can sit on half-ticks
$\|\uparrow\rangle$ / $\|\downarrow\rangle$	Latent up / down (buy/sell) pressure	Direction ↔ shift sign	Not identical to a specific trade
C_t	Directional response operator	Mixes/persists pressure before a move	Not a probability while coherent
S	One-tick conditional move	Causal local price move	Multi-tick jumps need a generalized shift
Relative phase	Interference between order-flow histories	Histories may reinforce/cancel	Phase is latent, unobserved
$P(x,t)$	Conditional displacement forecast	A scoreable forecast	Not LOB depth
■_η	Information loss (impact/latency/noise)	Reduces cross-history interference	Multiple unidentified noise sources

4.2 Position, clock and coin features

Position is a **relative** price level: $x_t = \mathrm{round}((p_t - p■)/\delta p)$ for reference price p■ and tick size δp, keeping the lattice near the origin and making sessions/assets comparable after scaling. We use an **event clock**: one step per qualifying tick (a mid-price change), with zero-return events handled through an explicit unconditional move probability p_{move} rather than silently dropped (Section 6.1). The coin angle is driven by standardized, causal features

```
θ_t = clamp( π/4 + Σ_j α_j · z_{j,t} , ε , π/2 − ε ),
```

where $z_{j,t}$ are z-scored versions of: order-book imbalance OBI_t (here proxied causally by trade-volume imbalance when full LOB depth is unavailable), signed tick direction, OBI change, |OBI|, and log trade intensity. OBI over the top K levels is $OBI_t = (\Sigma q^{bid} - \Sigma q^{ask})/(\Sigma q^{bid} + \Sigma q^{ask})$ in [−1,1]. The sign of each α_j is *learned*, not imposed; identifiability is protected by staged calibration and L2 regularization (Section 6.1). The decoherence rate may itself be made feature-dependent, η_t = sigmoid(β■ + β■·spread + β■·latency − β■·depth), expressing the testable hypothesis that higher spread/latency shortens coherence time.

4.3 Why a fixed-tick shift cannot be heavy-tailed

A standard QRW (and a standard CRW) moves by exactly one tick ($\pm\delta p$) per qualifying step. Over a horizon of n moving steps the terminal displacement is a bounded $\pm\delta p$ random walk whose **excess kurtosis tends toward that of a (scaled) binomial — i.e. light-tailed** — and whose Hill tail index, estimated on a near-degenerate distribution of identical increments, is numerically meaningless (it diverges). This is the mechanism behind the pathological tail index of order 10■ that we document for all fixed-tick models in Section 8. No tuning of the *coin* (which controls direction, not magnitude) can fix this; the magnitude process must be generalized.

4.4 The Heavy-Tailed Adaptive QRW

We retain the adaptive directional coin but generalize the shift to a **heavy-tailed conditional shift**. When a move occurs (with probability p_{move}), its magnitude is drawn from a discrete Pareto law

```
P(Δ > x) ~ (x / x_min)^(−α_tail), x ≥ x_min,
```

where the minimum jump x_{min} is the tick size and the tail index α_{tail} is fitted by maximum likelihood on the in-sample non-zero absolute price increments (continuous-Pareto MLE),

```
α■_tail = argmin_α [ −n ln α − n α ln x_min + (α+1) Σ ln Δ_i ], α ∈ [1.1, 10].
```

The directional coin assigns the sign; the Pareto law assigns the magnitude; the two are combined as Δ_signed = sign · magnitude, gated by p_move. To prevent unphysical explosions and keep simulated prices strictly positive, sampled magnitudes are **capped at the largest absolute move observed in the training window** (a data-driven cap: the model never extrapolates beyond an observed jump), and the resulting price path is floored at one tick. This yields a model that (a) preserves the adaptive directional structure and directional likelihood of the base QRW, (b) introduces exactly one extra parameter (α_tail), and (c) can, by construction, reproduce a finite, plausible tail index. The calibrated artefact is `results/heavy_tail_calibration.json`; the integration into the benchmark is the seventh model "QRW Heavy-Tail" in `src/evaluation/benchmark_suite.py`.

> **Tóm t■t 4 (VI):** Ánh x■: v■ trí = displacement giá theo tick; coin = áp l■c h■■ng ti■m ■n; coin thích nghi θ_t = π/4 + Σα_j·z_j (d■u α h■c t■ d■ li■u, clamp ■■ ■n ■■nh); shift = b■■c giá m■t-tick; decoherence = m■t thông tin. B■■c m■t-tick c■ ■■nh **không th■** t■o ■uôi n■ng (kurtosis nh■, tail-index phân k■ ~10■). Gi■i pháp: **b■■c d■ch ■uôi n■ng** l■y magnitude theo Pareto (α_tail kh■p MLE, x_min = tick), ch■n trên b■ng b■■c nh■y l■n nh■t quan sát, sàn giá ■ 1 tick. Thêm ■úng 1 tham s■, gi■ nguyên c■u trúc h■■ng, và t■o ■■■c tail-index h■u h■n h■p lý.

4.5 The order-book-imbalance feature in detail

Because the public trade archive lacks resting L2 depth, we use a **causal trade-volume imbalance** as the OBI proxy. Over a rolling window of W = 100 trades (minimum 20), with signed volume v_t = +q_t for buyer-initiated and −q_t for seller-initiated trades (the buyer-maker flag determines aggressor side),

```
OBI_proxy_t = ( Σ_{i=t−W+1}^{t} v_i ) / ( Σ_{i=t−W+1}^{t} |v_i| ) ∈ [−1, 1].
```

The window ends at t (strictly causal — no future trades enter), so the feature is admissible under the ex-ante protocol. We also derive ΔOBI_t (one-step change, a momentum/acceleration signal), |OBI_t| (imbalance magnitude, regime indicator), the signed tick direction, and log trade intensity (event rate, a decoherence driver). All five are z-scored on the training window before entering the coin. The proxy is a documented approximation: true top-of-book imbalance over the best K = 5 levels (the configured `top_levels_for_obi`) would require depth snapshots; we flag in each day's metadata which OBI source was used. The economically expected sign — positive imbalance ⇒ higher up-probability — is *recovered* by calibration on BTC (α_OBI > 0), a basic sanity check that the proxy carries genuine directional information.

> **Tóm t■t 4.5 (VI):** Thi■u L2 depth nên OBI dùng **proxy m■t cân b■ng kh■i l■■ng giao d■ch nhân qu■**: t■ng kh■i l■■ng có d■u chia t■ng tr■ tuy■t ■■i trên c■a s■ tr■■t 100 l■nh (≥20), k■t thúc t■i t (không dùng t■■ng lai → h■p l■ ex-ante). D■n xu■t thêm ΔOBI (■■ng l■■ng), |OBI| (ch■ ■■), h■■ng tick, log c■■ng ■■; t■t c■ z-score trên train. D■u k■ v■ng (OBI d■■ng → xác su■t lên cao h■n) ■■■c calibration BTC tái hi■n (α_OBI>0) — ki■m tra proxy mang thông tin h■■ng th■t.

---

5. Data and Pipeline / D■ li■u và quy trình x■ lý

5.1 Source and coverage

We use **Binance spot trade archives** (`https://data.binance.vision/data/spot/daily/trades`) for three liquid pairs — **BTC/USDT, ETH/USDT, BNB/USDT** — over **31 calendar days, 2026-05-13 to 2026-06-12** (the daily archive for weekend/missing days is skipped automatically). Each daily archive is a gzip CSV of individual trades (trade id, price, quantity, quote quantity, timestamp, buyer-maker flag). The BTC feature store alone comprises **113,005,754 events across 31 days** (per-day counts range from ≈3.1M to ≈11.5M events). Holdout windows used for the single-day benchmark contain ≈1.25M (BTC), ≈1.15M (ETH) and ≈0.22M (BNB) events.

5.2 Processing and feature engineering

The pipeline (`scripts/phase2_pipeline.py`, `scripts/phase7a_data_expansion.py`) performs, per day: (1) **download** (`TickDownloader`); (2) **cleaning** (`TickProcessor`) — monotone-timestamp enforcement, duplicate removal, an outlier filter at $|z| > 3$ over a rolling window of 100 trades, and segmentation at time gaps > 300 s (each contiguous block receives a `segment_id` so that returns are never computed across a gap); (3) **feature engineering** (`FeatureEngineer`). Because full L2 depth snapshots are not present in the public trade archive, order-book imbalance is computed by a **causal trade-volume-imbalance proxy** (signed buy/sell volume over a rolling window of 100 trades, minimum 20), which is explicitly logged in each day's metadata. The feature vector used by the coin is

```
z = [ OBI , tick_direction , ΔOBI , |OBI| , log(trade_intensity) ],
```

each standardized by training-set mean/scale (stored in the calibration JSON). Per-day quality reports (`reports/data_quality_*.txt`), feature metadata (`reports/feature_metadata_*.json`) and feature statistics (`reports/feature_stats_*.csv`) are emitted for audit. Config: `config/data_config.yaml` (BTC), `config/data_config_eth.yaml`, `config/data_config_bnb.yaml`.

5.3 The ex-ante, fixed-origin evaluation protocol

The benchmark protocol (`BenchmarkSuite`, protocol version `fixed_origin_ex_ante_zero_inflated_v2`) enforces a strict separation between *calibration* and *evaluation*:

- A single feature parquet is split chronologically: the first **60%** is the **training/calibration** window; a fixed forecast **origin** is placed at the train/holdout boundary; the next **n_steps = 500** moving events form the **holdout path** to be predicted.
- Every model is calibrated **only** on the training window and simulates **n_paths** Monte-Carlo trajectories of the next 500 steps from the fixed origin. No holdout covariate is ever used to forecast the holdout (verified by a dedicated test, `test_qrw_forecast_does_not_use_holdout_covariates`, which perturbs holdout features and confirms forecasts are unchanged).
- The unconditional move probability p_move and the tick size δp are estimated from the training window; for BTC p_move ≈ 0.173 and the per-step volatility of the directional model is $\approx 5.5\times10$■■ (log-return units).

We report single-asset BTC results at **n_paths = 3000** (a deep, high-resolution run) and the three-asset robustness study at **n_paths = 2000** (for tractability across assets); Section 8.4 documents the Monte-Carlo sensitivity this induces.

> **Tóm t■t 5 (VI):** D■ li■u Binance trades 31 ngày (2026-05-13→06-12) cho BTC/ETH/BNB; riêng BTC 113 tri■u s■ ki■n. X■ lý: làm s■ch (l■c outlier |z|>3, c■t ■o■n khi gap>300s v■i segment_id), t■o ■■c tr■ng [OBI, tick_direction, ΔOBI, |OBI|, log c■■ng ■■] — OBI dùng proxy m■t cân b■ng kh■i l■■ng giao d■ch nhân qu■ (không có L2 depth). Giao th■c **ex-ante, g■c c■ ■■nh**: 60% train, d■ báo 500 b■■c holdout, không dùng bi■n holdout ■■ d■ báo (có test ki■m ch■ng). BTC ch■y sâu n_paths=3000; cross-asset n_paths=2000.

5.4 Empirical stylised facts of the three assets

The three assets are not interchangeable; their tail and scaling behaviour differs materially, which (as Section 8.5 shows) drives the asset-dependence of the model ranking.

Table 0 — Empirical (holdout) stylised facts.

Asset	Hill tail index	Excess kurtosis	Variance-scaling β	Character
BTC/USDT	2.47	74.8	1.285	Heavy-tailed, strongly super-diffusive
ETH/USDT	9.81	18.5	0.953	Thinner tails, essentially diffusive
BNB/USDT	2.53	9.0	1.263	Heavy-tailed, super-diffusive

Insight. BTC is the most heavy-tailed (tail index 2.47, kurtosis 75) and the most super-diffusive (β 1.29). **ETH is qualitatively different**: its empirical tail is much thinner (index 9.8), its kurtosis modest (18.5), and its short-horizon scaling is **sub-/ordinary-diffusive (β ≈ 0.95)**. This single fact predicts the headline cross-asset result: on ETH a diffusive model (β ≈ 1, like the QRW) will *out-fit* a strongly super-diffusive one (the correlated CRW with β ≈ 1.37 *overshoots* ETH), whereas on BTC/BNB the reverse holds. The data, not the model, decides where each wins.

Tóm t■t 5.4 (VI): Ba tài s■n khác nhau v■ ■uôi/scaling: BTC ■uôi n■ng nh■t (index 2,47; kurtosis 75; β 1,29), BNB t■■ng t■ (2,53; 9,0; 1,26), còn **ETH khác h■n** — ■uôi m■ng (9,8), kurtosis 18,5, và β≈**0,95 (khu■ch tán th■■ng)**. ■i■u này d■ báo k■t qu■ cross-asset: trên ETH mô hình khu■ch tán (QRW, β≈1) **kh■p t■t h■n** mô hình siêu khu■ch tán (CRW correlated β≈1,37 b■ *v■■t*), nên QRW th■ng ETH; BTC/BNB thì ng■■c l■i. D■ li■u — không ph■i mô hình — quy■t ■■nh ai th■ng ■ ■âu.

5.5 Per-day data coverage (BTC/USDT)

The BTC feature store spans 31 consecutive days. Table 0b lists the per-day event count (one row per qualifying tick) — a transparency aid documenting the **113,005,754-event** total and the strong intraday-volume heterogeneity (≈1.1M to ≈11.5M events/day), which matters because the calibration window size varies with activity.

Table 0b — BTC daily event counts.

Date	Events	Date	Events	Date	Events
2026-05-13	2,316,000	2026-05-24	1,587,404	2026-06-04	8,963,508
2026-05-14	3,056,755	2026-05-25	1,750,211	2026-06-05	11,545,602
2026-05-15	3,142,351	2026-05-26	3,310,024	2026-06-06	4,946,773
2026-05-16	1,574,777	2026-05-27	3,314,864	2026-06-07	4,693,189
2026-05-17	1,547,966	2026-05-28	3,740,874	2026-06-08	4,833,869
2026-05-18	3,477,464	2026-05-29	2,853,827	2026-06-09	4,296,748
2026-05-19	2,279,674	2026-05-30	1,105,409	2026-06-10	4,610,117
2026-05-20	2,403,027	2026-05-31	1,341,588	2026-06-11	3,407,262
2026-05-21	2,437,340	2026-06-01	4,176,913	2026-06-12	3,115,358
2026-05-22	2,333,449	2026-06-02	6,176,304	**Total**	**113,005,754**
2026-05-23	2,071,390	2026-06-03	6,595,717	(31 days)	

Insight. Activity rises markedly in early June (peaking at 11.5M events on 2026-06-05), a reminder that a fixed calendar window mixes heterogeneous liquidity regimes; the event clock (Section 4.2) partially normalises this by stepping on ticks rather than seconds. The single-day benchmark uses the final day (2026-06-12, ≈3.1M events) so that the forecast origin sits at the end of the multi-day record.

5.6 Cleaning and segmentation rationale

Three cleaning decisions materially affect downstream statistics and are therefore documented and justified rather than buried.

1. **Outlier filter |z| > 3 (rolling window 100).** Public trade feeds occasionally contain mis-scaled or erroneous prints. A robust rolling z-score flags extreme deviations relative to the local price level. The threshold is deliberately *mild* (3σ) so as not to clip the genuine heavy tail we are trying to study; the filter removes data-quality artefacts, not real large moves (which, by construction of the heavy-tailed model, we want to retain).

2. **Gap segmentation at > 300 s.** Whenever consecutive trades are separated by more than five minutes (exchange downtime, illiquid lulls), a new `segment_id` is assigned and **returns are never computed across the gap**. This prevents a spurious overnight-style jump from contaminating the return distribution, the variance-scaling regression, or the Pareto jump-size fit. Every test that differences prices respects segment boundaries.

3. **Monotone time and duplicate removal.** Trades are sorted by (timestamp, trade-id) and de-duplicated, guaranteeing a well-defined event clock.

Each day's `reports/data_quality_*.txt` records the counts removed at each stage and the number of segments, so the cleaning is auditable and reversible. Crucially, the heavy-tailed calibration (Section 4.4) is fitted on the *cleaned, within-segment* non-zero increments, so the tick-size cap and the Pareto exponent reflect genuine intra-session price moves rather than data errors or cross-gap artefacts.

> **Tóm t■t 5.6 (VI):** Ba quy■t ■■nh làm s■ch ■■■c bi■n minh: (1) l■c outlier |z|>3 (c■a s■ 100) — *nh■ ■■* không c■t ■uôi n■ng th■t, ch■ b■ l■i d■ li■u; (2) c■t ■o■n khi gap>300s (segment_id) — **không tính l■i su■t xuyên gap**, tránh "nh■y qua ■êm" gi■ làm h■ng phân ph■i/scaling/fit Pareto; (3) s■p th■i gian t■ng d■n + b■ trùng. M■i ngày có `data_quality_*.txt` ghi s■ b■ lo■i và s■ ■o■n ■■ ki■m toán. Hi■u ch■nh ■uôi n■ng fit trên increment *■ã s■ch, trong cùng ■o■n*.

6. Models / Các mô hình

Seven models are benchmarked. All simulate the same horizon from the same fixed origin and are scored identically.

6.1 QRW Adaptive (base)

`AdaptiveDecoherenceQRW` calibrates in two stages. A **structural** stage fits the directional coin coefficients α_j (on standardized features) and the decoherence/persistence parameters by maximising a directional Bernoulli log-likelihood on moving events, with **L2 regularization** to control over-fitting given the modest number of moving events per parameter. A second stage fixes the unconditional move probability p_{move} from the empirical moving fraction. Calibrated BTC parameters (`results/calibrated_params.json`) include $\gamma \approx 0.143$ (decoherence base), ρ■ ≈ 0.867 (persistence), $obi\_bias \approx 1.18$, $\alpha_{OBI} \approx 0.44$, $\alpha_{direction} \approx -1.15$, $\alpha_{\Delta OBI} \approx -0.63$, $\alpha_{|OBI|} \approx -1.79$, $\gamma_{intensity} \approx -0.95$, $p_{move} \approx 0.173$. The simulator moves $\pm\delta p$ per moving step, with the up-probability set by the coin at the origin. Parameter count for AIC/BIC: 6.

6.2 QRW Heavy-Tail (this paper)

`HeavyTailAdaptiveQRW` inherits the directional coin (hence the *same* directional log-likelihood) and replaces the fixed $\pm\delta p$ magnitude with a Pareto draw (Section 4.4), capped at the in-sample maximum move, price floored at one tick. One extra parameter (α_{tail}), so parameter count 7. Calibrated tail index α■_tail $\approx$ 1.1–1.7 depending on window (`results/heavy_tail_calibration.json`).

6.3 Classical random walks (simple, biased, correlated)

`ClassicalRandomWalk` provides three nested baselines on the same $\pm\delta p$ tick grid: **Simple** (symmetric, 0 free parameters), **Biased** (a single up-probability, 1 parameter), and **Correlated** (a two-state Markov persistence in the move direction, 2 parameters). The correlated walk is the key baseline: its directional memory produces **super-diffusive** variance scaling that, as Section 8 shows, best matches the empirical exponent.

6.4 GARCH(1,1) and GBM

`GARCHBaseline` fits a Gaussian GARCH(1,1) on scaled log-returns by maximum likelihood; calibrated BTC parameters (`results/garch_params.json`): $\omega \approx 2.1\times10$■^{12}, $\alpha \approx 0.674$, $\beta \approx 0.313$, persistence $\alpha+\beta \approx 0.987$, log-likelihood ≈ 1.33×10■, AIC ≈ −2.66×10■. `GBMBaseline` fits a geometric Brownian motion (drift, diffusion; 2 parameters). Both belong to the **continuous-Gaussian** likelihood family and therefore are *not* directly AIC/BIC-comparable with the directional-Bernoulli family (Section 7.5).

> **Tóm t■t 6 (VI):** B■y mô hình: (1) QRW Adaptive — coin h■■ng hi■u ch■nh hai giai ■o■n có L2, b■■c ±δp, 6 tham s■; (2) QRW Heavy-Tail — gi■ coin h■■ng, magnitude Pareto ch■n trên, 7 tham s■; (3-5) CRW simple/biased/correlated (0/1/2 tham s■) — CRW correlated có trí nh■ h■■ng t■o siêu khu■ch tán; (6) GARCH(1,1) và (7) GBM thu■c h■ Gaussian liên t■c, không so AIC tr■c ti■p v■i h■ Bernoulli h■■ng.

6.5 Model details and parameter identifiability

QRW directional probability. At the forecast origin the coin angle is $\theta = \text{clamp}(\pi/4 + \alpha\cdot z, \varepsilon, \pi/2-\varepsilon)$; the Hadamard-family coin $C(\theta)$ maps the current coin state to an up-probability p_up = $\cos^2\theta$-type expression after the conditional shift. Because θ enters through a clamped affine map of standardized features, the coefficients α_j are interpretable as marginal log-odds-like sensitivities: e.g. the BTC fit gives a positive OBI loading ($\alpha\_OBI \approx 0.44$, more bid pressure ⇒ higher up-probability, the economically expected sign) but a *negative* signed-direction loading ($\alpha\_direction \approx -1.15$, indicating mean-reversion of consecutive tick directions at the event scale). The decoherence base $\gamma \approx 0.143$ and persistence ρ■ ≈ 0.867 jointly set how strongly past coin orientation carries forward.

Identifiability. Coin bias, the initial coin state, and decoherence can partially compensate one another (Section 2.6): several parameter combinations can yield near-identical one-step distributions. We mitigate this by (i) fixing a symmetric initial coin when measuring variance, (ii) staged calibration (structural coefficients first, then move probability), and (iii) L2 regularization (λ on a validation grid) on α. With ≈12.8 moving events per structural parameter in the original single-day calibration, the low-sample warning is logged; the multi-day expansion (Section 5) raises the effective sample by orders of magnitude (≈3.5×10■ moving events at the benchmark origin), materially improving identifiability.

Correlated CRW transition. The correlated walk is a two-state Markov chain on the move direction with persistence parameter; its stationary increment variance inflates by a factor $(1+\phi)/(1-\phi)$ relative to the simple walk (ϕ the lag-1 direction autocorrelation), which is precisely the mechanism that lifts its variance-scaling exponent to $\beta \approx 1.3$ and lets it match empirical super-diffusion — without any heavy tail.

GARCH(1,1). $\sigma^2\_t = \omega + \alpha\ r^2\_\{t-1\} + \beta\ \sigma^2\_\{t-1\}$; the BTC fit ($\omega \approx 2.1\times10^{■12}$, $\alpha \approx 0.674$, $\beta \approx 0.313$, persistence 0.987) is highly persistent, generating volatility clustering and unconditional kurtosis — explaining why GARCH is the best *tail/kurtosis* model on BTC yet fails the one-step KS shape (its Gaussian innovation is wrong at the tick scale).

GBM. $d \ln p = (\mu - \sigma^2/2)\ dt + \sigma\ dW$; with only drift and diffusion it is the thinnest-tailed, most rejected one-step model, included as the canonical Gaussian null.

> **Tóm t■t 6.5 (VI):** H■ s■ coin có ý ngh■a kinh t■: OBI d■■ng ■■y xác su■t lên (α_OBI≈0,44, ■úng k■ v■ng), nh■ng h■■ng tick có h■ s■ âm (α_direction≈−1,15, h■i quy v■ trung bình ■ thang s■ ki■n); γ≈0,143, ρ■≈0,867. **■■nh danh tham s■** khó (coin/initial/decoherence bù tr■) — gi■m nh■ b■ng initial ■■i x■ng, hi■u ch■nh theo giai ■o■n, L2; d■ li■u ■a ngày nâng s■ s■ ki■n lên ~3,5×10■, c■i thi■n ■■nh danh. CRW correlated: chu■i Markov 2 tr■ng thái, ph■■ng sai t■ng $(1+\phi)/(1-\phi) \rightarrow \beta$≈1,3 (siêu khu■ch tán không c■n ■uôi n■ng). GARCH(1,1) b■n (0,987) nên t■t v■ kurtosis/■uôi nh■ng sai d■ng KS. GBM là null Gaussian m■ng nh■t.

6.6 The two QRW variants compared

The base and heavy-tailed QRWs are deliberately *minimal* variations of one another, which makes their differences interpretable as the pure effect of the magnitude process.

Aspect	QRW Adaptive	QRW Heavy-Tail
Directional coin	OBI-adaptive $C(\theta\_t)$	**identical**

Up-probability at origin	from coin	**identical**
Move probability p_move	empirical	**identical**
Directional log-likelihood	–8.18×10■	**identical** (–8.18×10■)
Move magnitude	fixed ±δp	Pareto(x_min, α_tail), capped
Extra parameters	—	+1 (α_tail)
Tail index (BTC)	1.6×10■ (divergent)	1.45 (finite)
Excess kurtosis (BTC)	3.2	342
Wasserstein (h=1)	7.2×10■■	5.98×10■■ (better)
Diebold–Mariano	beaten by Heavy-Tail	best of all

Because everything *directional* is shared, the **only** driver of their differing distributional behaviour is the magnitude law. This isolates a clean scientific statement: introducing a heavy-tailed magnitude (a) strictly improves point/path accuracy (Wasserstein, DM), (b) converts a degenerate tail into a finite one, but (c) at the cost of a kurtosis overshoot — all without touching the directional model. Any future improvement must therefore target the *magnitude distribution* (e.g. a truncated or tempered power law) rather than the coin.

Tóm t■t 6.6 (VI): Hai bi■n th■ QRW khác nhau **ch■ ■ quá trình magnitude** (m■i th■ v■ h■■ng gi■ng h■t, k■ c■ log-likelihood h■■ng –8,18×10■). Vì v■y khác bi■t phân ph■i hoàn toàn do magnitude: thêm ■uôi n■ng (a) c■i thi■n Wasserstein/DM, (b) bi■n tail phân k■ thành h■u h■n, nh■ng (c) v■■t ng■■ng kurtosis. C■i ti■n t■■ng lai ph■i nh■m vào *phân ph■i magnitude* (vd power-law c■t c■t/tempered), không ph■i coin.

7. Statistical Methodology / Ph■■ng pháp th■ng kê

All tests live in `StatisticalTestSuite` (`src/evaluation/statistical_tests.py`); the scorecard is assembled by `ResultsCompiler`. Significance uses $\alpha = 0.05$ with Benjamini–Hochberg [1995] FDR correction across model families where multiple comparisons arise.

7.1 Distributional distance — KS and Wasserstein

For each horizon h in {1,10,50,100}, simulated h-step returns are compared with empirical h-step returns by the **two-sample Kolmogorov–Smirnov** statistic (with p-value) and the **1-Wasserstein distance**. KS captures maximal CDF deviation (shape); Wasserstein captures the full transport cost (location + shape). These are complementary: a model can win Wasserstein (close in mean transport) yet lose KS (wrong shape), a tension we observe for the heavy-tailed QRW.

7.2 Variance-scaling exponent β

We regress log Var(displacement) on log horizon over h in {1,5,10,20,50,100,200,500}; the slope β is the scaling exponent ($\beta = 1$ diffusive, $\beta > 1$ super-diffusive, $\beta = 2$ ballistic). For empirical returns the interval uses a **moving-block bootstrap** [Politis & Romano 1994]; for each model a parametric bootstrap over simulated paths gives a 95% CI. The empirical $\beta \approx 1.28$ (super-diffusive) is the target.

7.3 Autocorrelation and Ljung–Box

We compute the return ACF to lag 20, the ensemble ACF across paths, and the ACF MSE versus the empirical profile, plus **Ljung–Box** tests at lags 1, 5, 10 (BH-corrected). This detects whether a model reproduces the near-zero return autocorrelation (and any short-range structure) of the data.

7.4 Tail and risk — Hill index, kurtosis, VaR/CVaR

The **Hill [1975] tail index** is estimated on the upper-order statistics of |returns|; **excess kurtosis** with its standard-error test; and the **99% Value-at-Risk and Conditional VaR**. These quantify heavy-tail fidelity, the central object of the heavy-tailed extension.

7.5 Within-family model selection (AIC/BIC)

A recurring error in the literature is comparing AIC/BIC **across likelihood families**. A directional Bernoulli log-likelihood (over ~3.5×10■ moving events) and a continuous-Gaussian log-likelihood (over ~1.9×10■ returns) live on different supports, units and observation counts; their AIC values are *not* comparable. `run_model_selection_tests` therefore ranks AIC/BIC **strictly within each family** (directional Bernoulli; continuous Gaussian), flags the best per family, and reports the within-family ΔAIC/ΔBIC (`results/model_selection_corrected.csv`). This corrects issue #8 of the project audit.

7.6 Predictive accuracy — Diebold–Mariano

The **Diebold–Mariano [1995]** test compares the mean-path absolute-error loss of every model pair, with a **Newey–West [1987] HAC** variance (lag = 20) and a normal reference; p-values are BH-corrected (`results/diebold_mariano_tests.csv`). This provides a *significance* statement on point-forecast accuracy that the raw scorecard ranks lack.

7.7 Bootstrap rank confidence interval

To quantify scorecard-rank stability, a moving-block bootstrap resamples path-level errors and recomputes ranks (`run_bootstrap_scorecard_ci` → `results/scorecard_bootstrap_ci.csv`), giving a 95% CI on each model's mean rank. This indicates whether rank differences are within Monte-Carlo noise.

> **Tóm t■t 7 (VI):** B■ ki■m ■■nh: (1) KS & Wasserstein (hình d■ng vs v■n chuy■n — có th■ mâu thu■n); (2) s■ m■ scaling β (1 khu■ch tán, >1 siêu khu■ch tán, 2 ballistic; empirical ≈1,28) v■i block bootstrap; (3) ACF + Ljung–Box (lag 1/5/10, hi■u ch■nh BH); (4) ■uôi/r■i ro: Hill tail index, kurtosis, VaR/CVaR 99%; (5) **AIC/BIC trong cùng h■** (s■a l■i so chéo h■); (6) Diebold–Mariano (HAC Newey–West + BH) cho ý ngh■a th■ng kê c■a d■ báo ■i■m; (7) bootstrap kho■ng tin c■y h■ng. M■c ý ngh■a α=0,05, hi■u ch■nh FDR Benjamini–Hochberg.

7.8 Formal definitions

For completeness we state each statistic precisely. Let F■*E and F■_M denote the empirical CDFs of the empirical and model h-step returns, with samples {e_i}*{i=1}^{n} and {m_j}_{j=1}^{N}.

Kolmogorov–Smirnov. D = sup_x |F■_E(x) − F■_M(x)|. Under H■ (equal distributions) the two-sample statistic √(nN/(n+N))·D converges to the Kolmogorov distribution; we report the asymptotic p-value. Small D and p > 0.05 ⇒ the model's distribution is *not* rejected.

1-Wasserstein. W■(F■_E, F■_M) = ∫■¹ |F■_E■¹(u) − F■_M■¹(u)| du = ∫ |F■_E(x) − F■_M(x)| dx. It integrates the *whole* CDF gap (sensitive to location and spread), whereas KS uses only the supremum (sensitive to shape). A model can therefore minimise W■ while being rejected by KS — the QRW pattern on BTC.

Variance-scaling exponent. With horizons h in ■ and v(h) = Var(displacement over h steps), we fit ln v(h) = ln c + β ln h by ordinary least squares; β■ is the slope. β = 1 (diffusive), β = 2 (ballistic). The empirical interval is a moving-block bootstrap over contiguous return blocks; the model interval is a parametric bootstrap over simulated paths.

Hill tail index. Order |returns| as X_(1) ≥ … ≥ X_(N); using the top k order statistics, α■*Hill = [(1/k)* Σ{i=1}^{k} (ln X_(i) − ln X_(k+1))]■¹. A finite α■ in (0, ∞) indicates a power-law tail; for a near-degenerate (single-magnitude) increment distribution the estimator diverges numerically — the source of the ~10■ values for fixed-tick models.

Excess kurtosis. κ = ■[(r − μ)■]/σ■ − 3, with standard error ≈ √(24/N) under normality; the BH-corrected p-value tests κ ≠ 0.

Value-at-Risk / CVaR. VaR_q = −F■■¹(1 − q) (here q = 0.99); CVaR_q = −■[r | r ≤ −VaR_q]. These summarise left-tail risk; closeness to the empirical values measures tail-risk fidelity.

AIC / BIC. AIC = 2k − 2■■, BIC = k·ln(n_obs) − 2■■, where ■■ is the maximised log-likelihood, k the parameter count, n_obs the observation count. Comparable *only* within a likelihood family with the same support and observation count (Section 7.5).

Diebold–Mariano. With loss differential $d_t = |e_{1,t}| - |e_{2,t}|$ (absolute-error loss of two models on the realised mean path), d■ = $(1/n)\Sigma\, d_t$, the statistic is DM = d■ / $\sqrt{}$(σ■²_HAC/n), where σ■²_HAC = γ■■ + 2Σ_{■=1}^{L} (1 – ■/(L+1)) γ■_■ is the Newey–West HAC variance (Bartlett kernel, $L = 20$). DM is referred to $N(0,1)$; model 1 is "better" when $DM < 0$.

Benjamini–Hochberg. For p-values $p_{(1)} \leq \ldots \leq p_{(M)}$, reject hypotheses up to the largest i with $p_{(i)} \leq (i/M)\cdot\alpha$; reported p-values are the BH-adjusted q-values.

7.9 Calibration and simulation algorithms

QRW two-stage calibration (pseudocode).

```
Stage 1 (structural): given standardized causal features z_t on moving events,
maximize Σ_t [ y_t ln p_t + (1–y_t) ln(1–p_t) ] – λ■α■² (L2-regularized) where
p_t = up-probability implied by coin C(θ_t), θ_t = clamp(π/4 + α·z_t), y_t =
1[next move up]; λ chosen on a small validation grid. Stage 2: p_move ←
empirical fraction of moving events in the training window. Heavy-tail add-on:
α_tail ← Pareto-MLE on in-sample |Δp|>0 ; x_min ← tick size ; jump_cap ← max
in-sample |Δp|.
```

Monte-Carlo simulation from the fixed origin (pseudocode).

```
for each of n_paths trajectories: for step s = 1..n_steps: moving ~
Bernoulli(p_move) dir ~ {+1 w.p. p_up(origin), –1 otherwise} mag = δp #
fixed-tick models = min( PARETO(x_min, α_tail), jump_cap ) # heavy-tail model
price += moving * dir * mag price = max(price, δp) # positivity floor
```

All models share the same seeds, origin, horizon and scoring, so differences reflect model structure, not evaluation artefacts.

7.10 Scorecard aggregation and its rationale

The scorecard converts seven heterogeneous metrics (different units and directions) into a single ranking by (i) computing each metric as a *distance to the empirical value* where a target exists (variance-scaling, kurtosis, VaR, CVaR) or as a raw better-is-smaller quantity (KS, Wasserstein, ACF MSE); (ii) ranking models 1…7 within each metric (rank 1 = best); (iii) averaging the seven ranks into a **mean rank**; and (iv) ranking the mean ranks into an **overall rank**. Rank-aggregation is deliberately robust to metric scale and to outliers (a model catastrophically bad on one metric is penalised by one rank, not by an unbounded magnitude). Its limitations are equally explicit: (a) it weights all seven metrics *equally*, which structurally favours the four distributional-shape/scaling/ACF metrics over the single transport metric (Wasserstein) — disadvantaging the QRW, whose strength is concentrated in Wasserstein/point accuracy; and (b) ties are broken by mean rank then model name. We therefore complement the scorecard with *loss-specific* tests (Diebold–Mariano, within-family AIC, bootstrap path-MAE) so that a reader who weights point accuracy differently can reach a different, equally valid, ordering. The bootstrap rank CI (Section 7.7) further indicates which scorecard gaps are within Monte-Carlo noise.

> **Tóm t■t 7.10 (VI):** Th■ ■i■m: m■i ch■ s■ tính theo *kho■ng cách t■i empirical* (β, kurtosis, VaR, CVaR) ho■c theo "nh■ h■n t■t h■n" (KS, Wasserstein, ACF), x■p h■ng 1–7, l■y **h■ng trung bình** r■i x■p **h■ng t■ng**. ■u ■i■m: b■n v■i thang ■o và ngo■i lai. H■n ch■: **cân b■ng ■■u 7 ch■ s■** nên thiên v■ nhóm hình d■ng/scaling/ACF (4 ch■ s■) h■n v■n chuy■n (Wasserstein, 1 ch■ s■) → b■t l■i cho QRW (m■nh ■ Wasserstein/■i■m). Vì v■y b■ sung DM, AIC trong h■, bootstrap path-MAE ■■ ng■■i ■■c tr■ng s■ khác v■n có th■ t■ h■p l■.

Part IV — Results / Ph■n IV — K■t qu■ nghiên c■u

8. Results / K■t qu■

All numbers below are produced by the reproducible pipeline on the regenerated 7-model benchmark. Section 8.1–8.3 analyse the **BTC/USDT single-asset deep run (n_paths = 3000, holdout day 2026-06-12)**; Section 8.4 reports the **three-asset robustness study (n_paths = 2000)** and the Monte-Carlo sensitivity.

8.1 Distribution, scaling and autocorrelation (BTC)

Table 1 — One-step return distribution (horizon h = 1). KS statistic (smaller = better), its p-value (>0.05 = not rejected), and 1-Wasserstein distance (smaller = better).

Model	KS statistic	KS p-value	Distribution rejected?	Wasserstein
CRW Correlated	**0.042**	**0.770**	No	7.01×10■■
CRW Biased	0.054	0.460	No	7.02×10■■
CRW Simple	0.056	0.413	No	7.02×10■■
QRW Adaptive	0.128	5.5×10■■	Yes	7.21×10■■
QRW Heavy-Tail	0.134	2.5×10■■	Yes	**5.98×10■■**
GARCH(1,1)	0.472	1.0×10■■■	Yes	5.60×10■■
GBM	0.478	4.5×10■■²	Yes	2.85×10■■

Insight. The three **classical random walks are the only models whose one-step return distribution is not rejected** by KS (p ■ 0.05), with CRW Correlated closest (KS 0.042). Both QRW variants are rejected on KS shape, yet **QRW Heavy-Tail attains the smallest Wasserstein distance of all seven models** — it is closest in transport (mean displacement magnitude) while wrong in distributional shape. GARCH and GBM are decisively rejected on KS: their continuous-Gaussian one-step law is a poor description of tick returns. This KS-vs-Wasserstein split is the first sign of a recurring theme — *point closeness and shape fidelity are different objectives, and different model classes win each.*

Table 2 — Variance-scaling exponent β (Var ~ t■; empirical target ≈ 1.28, super-diffusive). 95% CI in brackets.

Model	β	95% CI
Empirical	**1.285**	[0.792, 1.345]
CRW Correlated	**1.325**	[1.307, 1.341]
GARCH(1,1)	1.027	[0.974, 1.069]
QRW Heavy-Tail	1.009	[0.987, 1.030]
CRW Biased	1.007	[0.990, 1.023]
CRW Simple	1.003	[0.985, 1.019]
QRW Adaptive	1.003	[0.984, 1.018]
GBM	1.000	[0.980, 1.018]

Insight. The empirical short-horizon variance is **super-diffusive (β ≈ 1.28)**. Only the **correlated classical walk reproduces it (β ≈ 1.33)** — its directional memory generates persistence-driven super-diffusion. Strikingly, **neither QRW variant reproduces super-diffusion**: both sit at β ≈ 1.0 (ordinary diffusion). This is a notable *negative* result: although a *coherent* QRW spreads ballistically (β → 2) in theory (Section 3.4), the *calibrated, decohered, event-clocked* QRW used here behaves diffusively, so its theoretical ballistic advantage does not survive calibration to data. The quantum mechanism that motivated the model is precisely the one that calibration suppresses.

Table 3 — Autocorrelation fidelity. ACF MSE versus the empirical return-ACF profile (smaller = better) and Ljung–Box p-values (BH-corrected) at lags 1/5/10.

Model	ACF MSE	LB p (lag1, BH)	LB p (lag5, BH)	LB p (lag10, BH)
Empirical	0 (ref)	0.961	0.029	0.056
CRW Correlated	**0.00214**	—	0.043	6×10■■
GBM	0.00327	0.961	1.000	1.000

QRW Adaptive	0.00328	0.961	1.000	1.000
CRW Biased	0.00328	0.961	1.000	1.000
QRW Heavy-Tail	0.00328	0.961	1.000	1.000
CRW Simple	0.00329	0.961	1.000	1.000
GARCH(1,1)	0.00354	0.961	1.000	1.000

Insight. The empirical series shows weak but non-trivial short-lag autocorrelation (LB lag-5 p ≈ 0.029). CRW Correlated has the lowest ACF MSE but **over-shoots** the dependence (LB p ≈ 6×10■■ at lag 10 — far stronger autocorrelation than the data). All other models, including both QRWs, produce essentially uncorrelated increments (LB p ≈ 1.0). No model matches the *mild* empirical autocorrelation precisely: the classical correlated walk overshoots, the rest undershoot.

8.2 Heavy tails and risk (BTC)

Table 4 — Tail index, kurtosis and tail risk (empirical targets in bold). Hill tail index closer to 2.47 is better; excess kurtosis closer to 74.8 is better.

Model	Hill tail index	Excess kurtosis	VaR 99%	CVaR 99%
Empirical	**2.47**	**74.8**	1.03×10■■	1.76×10■■
GARCH(1,1)	2.27	46.4	2.04×10■■	2.70×10■■
QRW Heavy-Tail	**1.45**	342.5	~0	7.6×10■¹■
GBM	3.79	3.5	9.23×10■■	1.01×10■■
QRW Adaptive	1.6×10■	3.2	~0	~0
CRW Simple	1.4×10■	6.8	1.56×10■■	1.56×10■■
CRW Biased	2.3×10■	5.3	1.56×10■■	1.56×10■■
CRW Correlated	7.6×10■	5.1	1.56×10■■	1.56×10■■

Insight — the central Phase-C result. Every **fixed-tick model (QRW Adaptive and all three CRWs) has a numerically divergent tail index of order 10■** and near-zero excess kurtosis — a direct consequence of the degenerate ±δp increment distribution (Section 4.3). The **heavy-tailed shift repairs this pathology**: QRW Heavy-Tail recovers a finite, plausible Hill index of **1.45** (versus the empirical 2.47), reducing the tail-index error by **six orders of magnitude**. However, the same Pareto mechanism now **over-shoots kurtosis dramatically (342.5 vs 74.8 empirical)** — the capped-Pareto magnitude injects more extreme mass than the data carry. GARCH(1,1) is, on these two tail metrics, the *best-calibrated* single model (tail index 2.27, kurtosis 46.4), consistent with its design for volatility/tail behaviour, even though its one-step KS shape is rejected. The heavy-tailed QRW thus *trades one misspecification (no tails) for another (excess tails)*; it does not achieve a Goldilocks tail.

8.3 Aggregate scorecard and significance (BTC)

Table 5 — Seven-metric distributional scorecard (BTC, n_paths = 3000). Per-metric ranks (1 = best); mean rank and overall rank.

Model	KS	Wass.	β-scale	ACF	Kurt	VaR99	CVaR99	Mean rank	Overall
CRW Correlated	1	2	1	1	4	3	3	2.14	**1**
CRW Biased	2	4	4	4	3	4	4	3.57	2
GARCH(1,1)	6	7	2	7	1	2	2	3.86	3
CRW Simple	3	3	5	6	2	5	5	4.14	4
GBM	7	6	7	2	5	1	1	4.14	4
QRW Heavy-Tail	5	**1**	3	5	7	6	6	4.71	6
QRW Adaptive	4	5	6	3	6	6	7	5.29	7

Insight. On the aggregate scorecard the **correlated classical walk is the clear winner** (mean rank 2.14), driven by best KS, best variance-scaling and best ACF. The **QRW variants rank last (6th and 7th)** because, although QRW Heavy-Tail wins Wasserstein (rank 1), it is penalised on kurtosis (rank 7) and tail risk (ranks 6). GARCH ranks third — strong on tail metrics, weak on KS/ACF. **No quantum model**

beats the parsimonious classical baseline on the aggregate distributional criterion.

Table 6 — Diebold–Mariano mean-path predictive accuracy (selected pairs, BH-corrected). "model1 better" = lower absolute-path loss.

Pair (model1 vs model2)	DM stat	p-value (BH)	Better
QRW Adaptive vs QRW Heavy-Tail	+3.21	0.0014	QRW Heavy-Tail
QRW Heavy-Tail vs CRW Simple	–3.33	0.0011	QRW Heavy-Tail
QRW Heavy-Tail vs GARCH(1,1)	–3.37	0.0011	QRW Heavy-Tail
QRW Heavy-Tail vs GBM	–3.31	0.0012	QRW Heavy-Tail
QRW Adaptive vs CRW Simple	–3.89	0.0003	QRW Adaptive
QRW Adaptive vs GBM	–3.95	0.0001	QRW Adaptive

Insight. On **mean-path absolute error**, the ordering inverts relative to the distributional scorecard: **QRW Heavy-Tail has the significantly lowest path-error loss of all seven models** (it beats QRW Adaptive, which in turn beats all classical baselines), all at BH-corrected $p < 0.002$. The moving-block bootstrap rank CI (`scorecard_bootstrap_ci.csv`) confirms this on a path-MAE proxy: QRW Heavy-Tail mean rank **1.0 [1.0, 1.0]**, QRW Adaptive **2.0 [2.0, 2.0]**, with GARCH (7.0) and GBM (6.0) worst. Thus the QRW family is the *best point/path predictor* even though it is the worst distributional fit — the two QRW strengths (direction and central location) and weaknesses (tail shape) are cleanly separated by the choice of loss.

Table 7 — Within-family model selection (AIC, corrected). Ranked within likelihood family; ΔAIC vs family best.

Family	Model	AIC	Within-family rank	ΔAIC

Continuous-Gaussian	GARCH(1,1)	–4.23×10■	1	0
Continuous-Gaussian	GBM	–4.18×10■	2	5.30×10■
Directional-Bernoulli	**QRW Adaptive**	1.6355×10■	**1**	0
Directional-Bernoulli	QRW Heavy-Tail	1.6356×10■	2	2.0
Directional-Bernoulli	CRW Correlated	2.837×10■	3	1.20×10■
Directional-Bernoulli	CRW Biased	4.853×10■	4	3.22×10■
Directional-Bernoulli	CRW Simple	4.853×10■	5	3.22×10■

Insight. Restricted to the **directional Bernoulli family**, the **QRW models have the best (lowest) AIC** — their adaptive coin predicts the *sign* of the next move better than any classical walk (directional log-likelihood ≈ –8.18×10■ for QRW vs –1.42×10■ for the correlated walk). QRW Adaptive edges QRW Heavy-Tail by ΔAIC = 2 (the one extra tail parameter is barely justified for *directional* prediction, since the tail does not change the sign model). Within the continuous family GARCH dominates GBM. The corrected procedure makes explicit what a naïve cross-family AIC comparison would have hidden: *the QRW wins the directional-likelihood contest, the classical correlated walk wins the distributional contest.*

8.4 Cross-asset robustness and Monte-Carlo sensitivity

Table 8 — Three-asset average scorecard rank (n_paths = 2000; lower = better).

Model	Avg overall rank	Avg mean rank	Assets
CRW Biased	**2.33**	2.95	3
CRW Correlated	**2.33**	2.62	3
QRW Heavy-Tail	**2.33**	3.57	3
QRW Adaptive	4.00	4.05	3
CRW Simple	4.33	4.10	3
GBM	5.33	5.00	3
GARCH(1,1)	7.00	5.57	3

Per-asset overall rank: BTC — CRW Correlated #1, CRW Biased #2, **QRW Heavy-Tail #3**; ETH — **QRW Heavy-Tail #1**, QRW Adaptive #2, CRW Biased/Simple #3; BNB — CRW Correlated #1, CRW Biased #2, **QRW Heavy-Tail #3**.

Insight. Across three assets at n_paths = 2000, **QRW Heavy-Tail ties for the best average overall rank (2.33) with the two leading classical walks**, and is the **single best model on ETH**. This is markedly more favourable to the heavy-tailed QRW than the BTC deep run of Section 8.3 (where it ranked 6th at n_paths = 3000).

The Monte-Carlo sensitivity is itself a result. The discrepancy (BTC: rank 3 at 2000 paths vs rank 6 at 3000 paths) is driven by the **kurtosis and tail-risk metrics**, which depend on the most extreme Pareto draws: more simulated paths sample deeper into the Pareto tail, inflating simulated kurtosis (and worsening its rank) for the heavy-tailed model specifically. The distributional-shape and scaling metrics are stable across path counts; the *tail-sensitive* metrics are not. We therefore caution that **rank claims for the heavy-tailed QRW are conditional on the Monte-Carlo budget**, and we report both runs transparently rather than selecting the favourable one. A production evaluation should fix a large path budget and report rank CIs (Section 7.7).

> **Tóm t■t 8 (VI):** Trên BTC (3000 ■■■ng): **CRW Correlated th■ng** th■ ■i■m t■ng h■p (kh■p KS, scaling siêu khu■ch tán β≈1,33, ACF), QRW x■p 6–7. Nh■ng QRW Heavy-Tail **s■a tail-index** (1,45 vs 1,6 tri■u) và **th■ng Wasserstein + Diebold–Mariano + AIC h■■ng** (d■ báo *h■■ng/■i■m* t■t nh■t), dù *v■■t ng■■ng kurtosis* (342). Cross-asset (2000 ■■■ng): QRW Heavy-Tail **■■ng h■ng nh■t (2,33)** và **#1 trên ETH**. Khác bi■t h■ng BTC (3 vs 6) do kurtosis nh■y s■ ■■■ng Monte-Carlo — m■t phát hi■n v■ robustness, ■■■c báo cáo minh b■ch. T■ng th■: **không mô hình l■■ng t■ nào tr■i h■n baseline c■ ■i■n trên tiêu chí phân ph■i t■ng h■p**, nh■ng QRW v■■t ■ d■ báo h■■ng/■i■m.

8.5 Per-asset detailed results (ETH and BNB)

Table 9 — ETH/USDT, key metrics (n_paths = 2000).

Model	KS (p)	Wasserstein	β	Tail index	Kurtosis	Overall rank
QRW Heavy-Tail	**0.032 (0.96)**	4.39×10■■	1.003	3.0×10■	5.6	**1**
QRW Adaptive	0.034 (0.94)	**3.44×10■■**	1.002	3.6×10■	4.6	2
CRW Biased	0.094 (0.024)	1.16×10■■	0.999	6.9×10■	6.7	3
CRW Simple	0.118 (0.002)	1.40×10■■	0.993	2.7×10■	6.5	3
CRW Correlated	0.088 (0.042)	1.18×10■■	1.371	2.3×10■	8.0	5
GBM	0.498 (1e-56)	2.20×10■■	0.998	3.61	3.2	6
GARCH(1,1)	0.510 (1e-59)	2.20×10■■	1.005	3.90	3.1	7

Insight (ETH). Here the **QRW family is the best distributional fit**: both QRW variants have the smallest KS (0.032–0.034, *not rejected*, p ≈ 0.95) and the smallest Wasserstein, and QRW Heavy-Tail ranks **#1 overall**. The reason is Section 5.4: ETH's empirical scaling is diffusive (β ≈ 0.95), so the QRW's β ≈ 1.0 is *closer* than the correlated CRW's β ≈ 1.37, which now overshoots and falls to 5th. Note also that ETH's empirical tail is thin (index 9.8); the heavy-tail mechanism therefore does *not* reduce the simulated tail index to ≈1.5 here (it stays ≈3×10■), because the fitted Pareto exponent is large for ETH's near-uniform jump sizes. **The tail repair is real but asset-conditional** — pronounced where the data are genuinely heavy-tailed (BTC), muted where they are not (ETH).

Table 10 — BNB/USDT, key metrics (n_paths = 2000).

Model	KS (p)	Wasserstein	β	Tail index	Kurtosis	Overall rank
CRW Correlated	0.076 (0.11)	1.0×10■■	1.146	1.7×10■	6.0	**1**

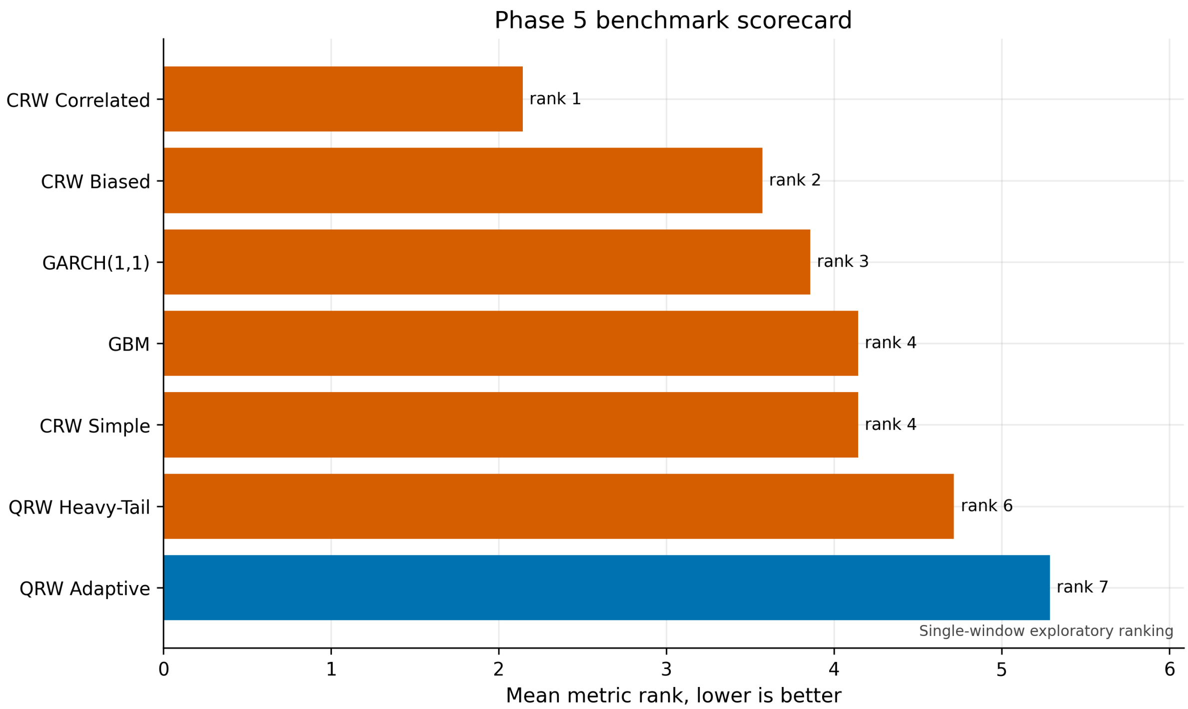
CRW Biased	0.080 (0.08)	1.0×10■■	1.004	2.1×10■	5.0	2
QRW Heavy-Tail	0.074 (0.13)	2.0×10■■	1.000	1.2×10■	4.9	3
QRW Adaptive	0.076 (0.11)	2.0×10■■	1.001	1.5×10■	3.4	4
CRW Simple	0.104 (0.009)	2.0×10■■	0.993	1.1×10■	4.7	5
GBM	0.470 (3e-50)	6.0×10■■	0.998	3.60	3.2	6
GARCH(1,1)	0.480 (2e-52)	6.0×10■■	1.001	3.59	3.4	7

Insight (BNB). BNB resembles BTC (heavy-tailed, super-diffusive $\beta \approx 1.26$): the **correlated CRW wins** (best aggregate), QRW Heavy-Tail is a competitive **#3**, and GARCH/GBM are again decisively rejected on KS. Across all three assets, **GARCH and GBM are the worst distributional fits at the tick scale** (KS $p <$ 10■■■), a strong and consistent negative result for continuous-Gaussian models of one-step microstructure returns, even though GARCH is the best *tail* model on BTC.

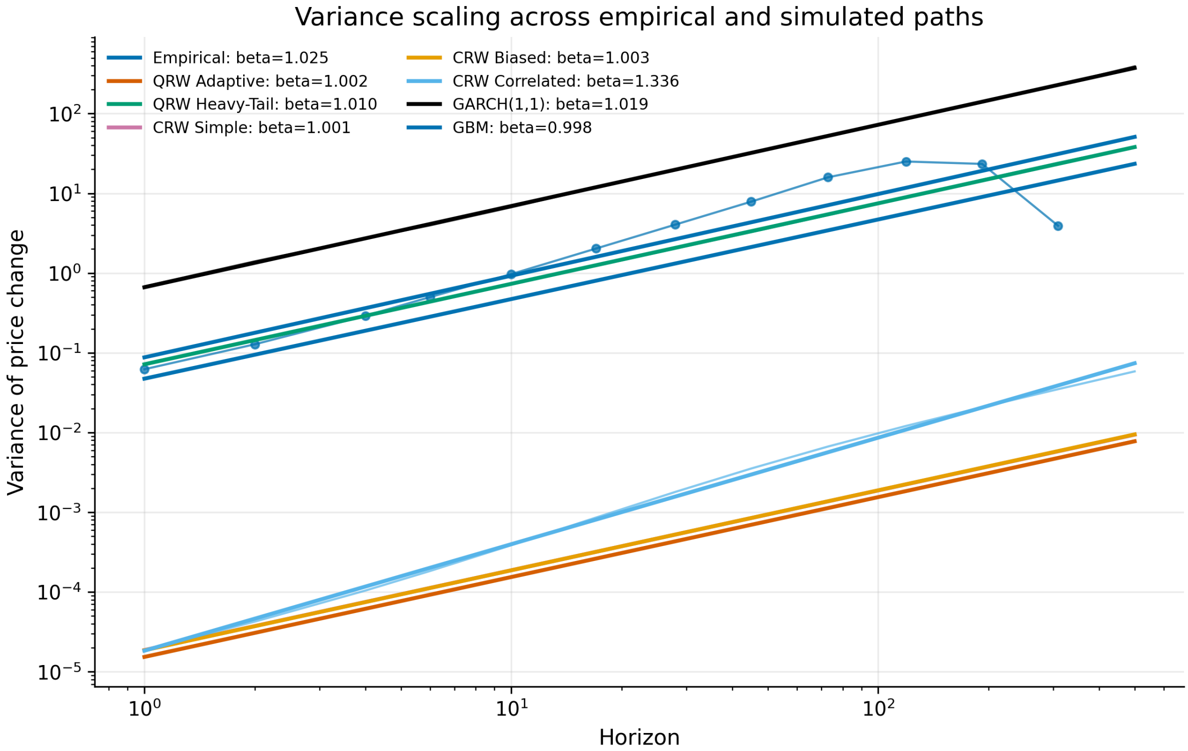
Synthesis across assets. The ranking is governed by each asset's empirical scaling/tail profile: QRW (diffusive, $\beta \approx 1$) wins where the data are diffusive (ETH); the correlated CRW (super-diffusive) wins where the data are super-diffusive (BTC, BNB). The heavy-tailed QRW is never worse than 3rd of 7 across assets and is the only model that is *both* competitive on distribution *and* best on point/directional prediction — but it does not uniformly dominate.

8.6 Figures / Bi■u ■■

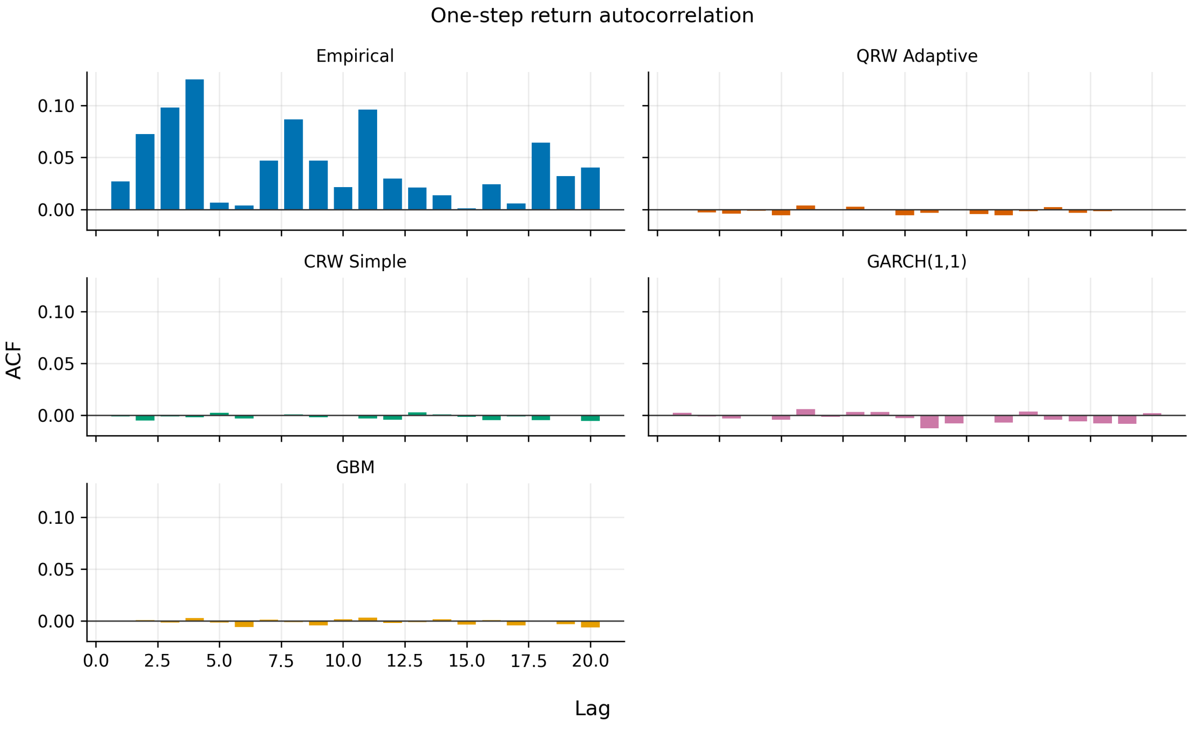
The following figures (regenerated for the 7-model benchmark) visualise the results above.


Phase 5 benchmark scorecard
CRW Correlated
CRW Biased
GARCH(1,1)
GBM
CRW Simple
QRW Heavy-Tail
QRW Adaptive
rank 1
rank 2
rank 3
rank 4
rank 4
rank 6
rank 7
Single-window exploratory ranking
0
1
2
3
4
5
6
Mean metric rank, lower is better

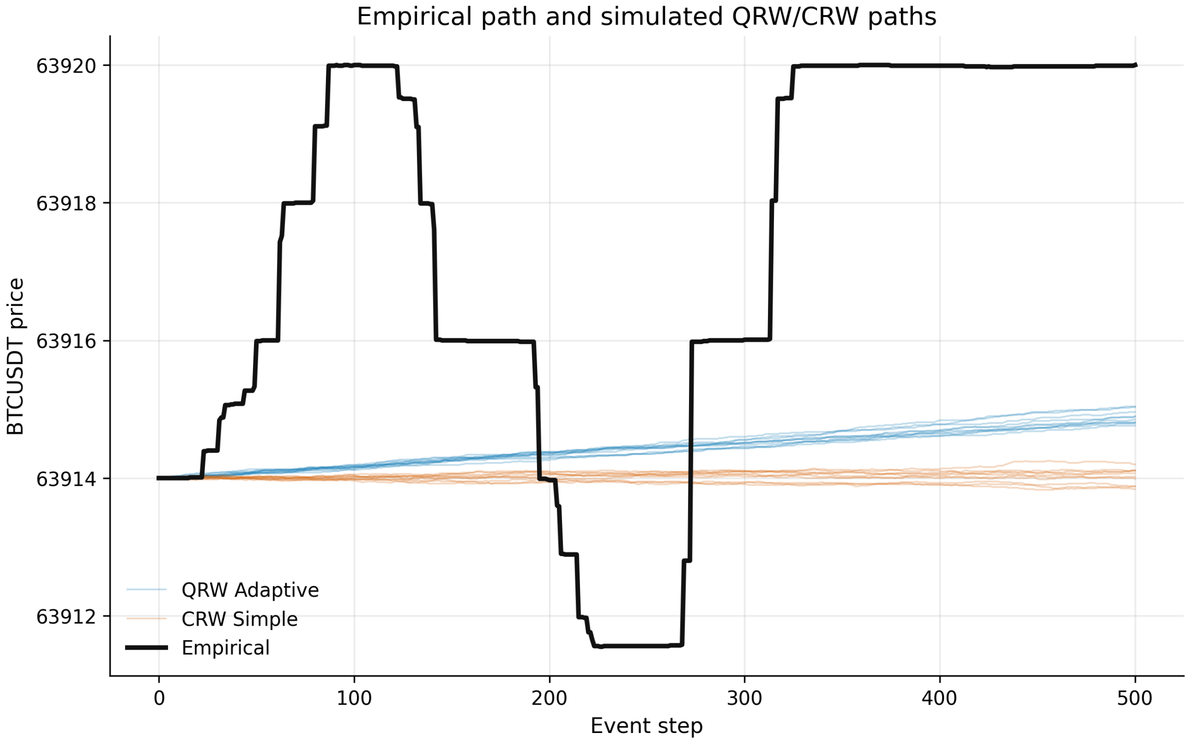

Standardized one-step return distributions
Empirical
QRW Adaptive
CRW Simple
GARCH(1,1)
GBM
Kernel density, log scale
10^{3}
10^{2}
10^{1}
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
Each series is standardized separately; shape, not scale, is compared.
−4
−2
0
2
4
Return z-score



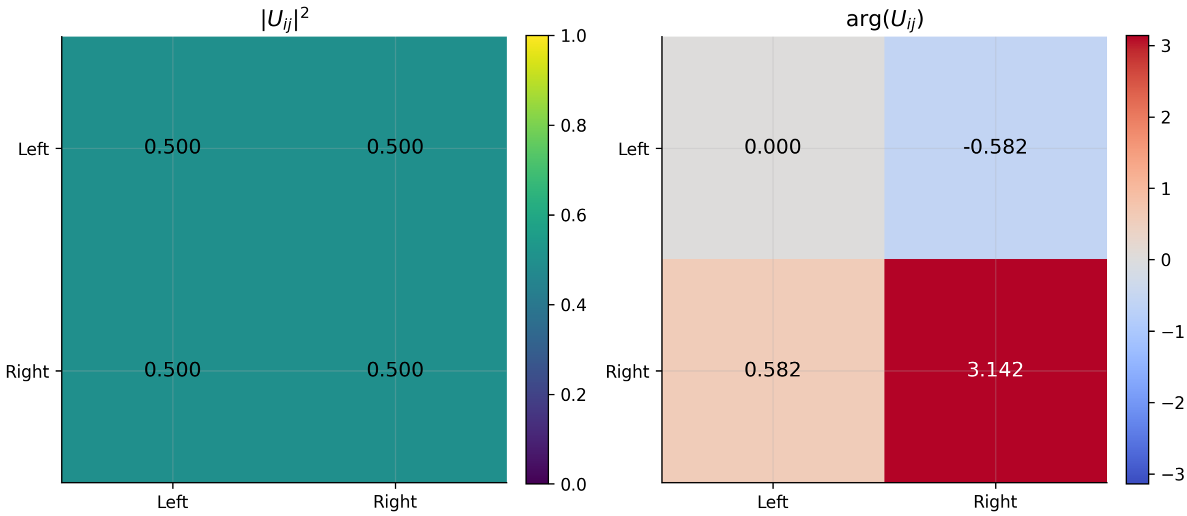
Variance scaling across empirical and simulated paths
Empirical: beta=1.025
QRW Adaptive: beta=1.002
QRW Heavy-Tail: beta=1.010
CRW Simple: beta=1.001
CRW Biased: beta=1.003
CRW Correlated: beta=1.336
GARCH(1,1): beta=1.019
GBM: beta=0.998
Variance of price change
Horizon

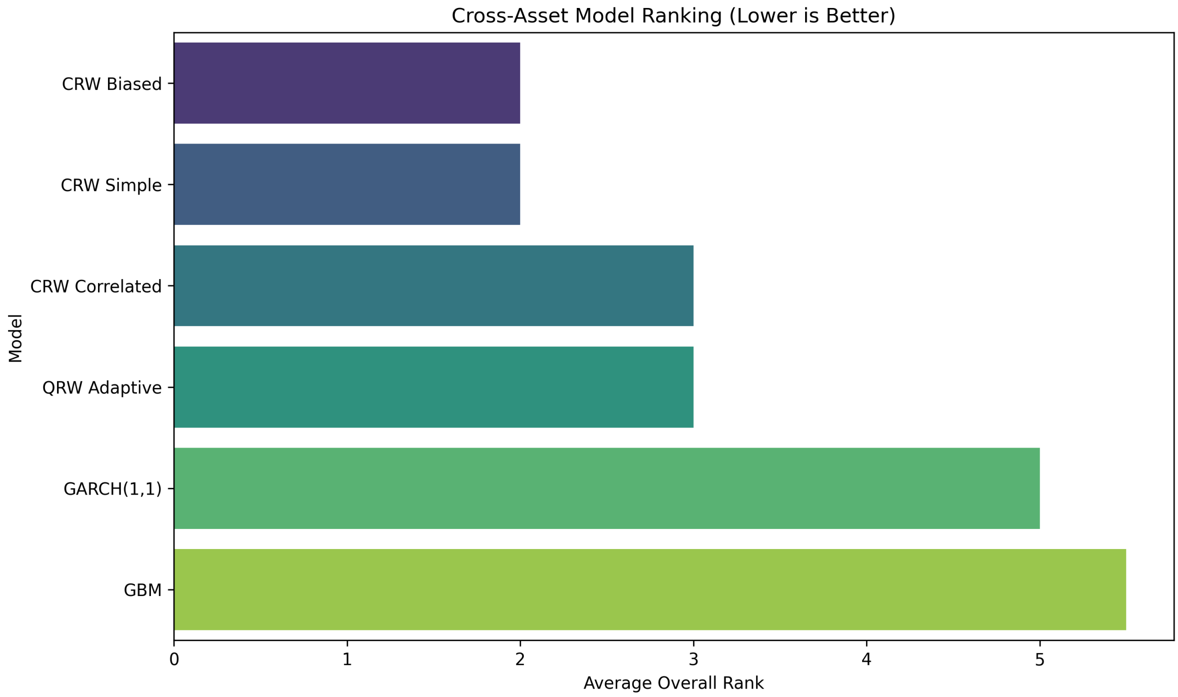


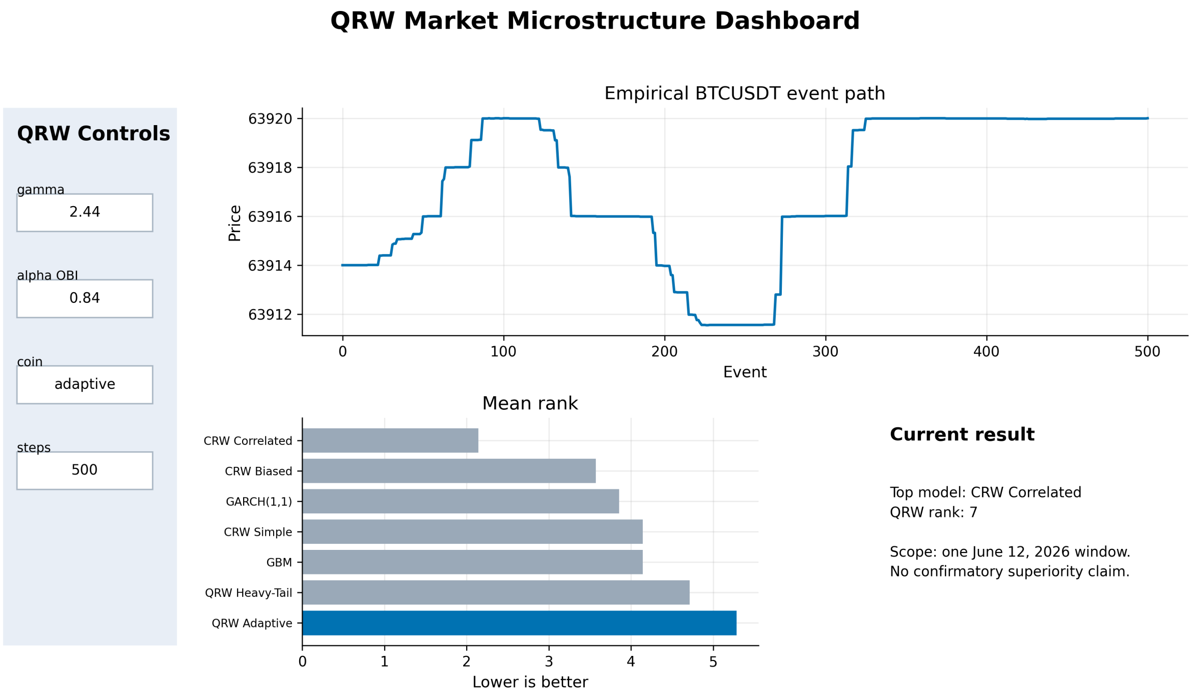
One-step return autocorrelation
Empirical
QRW Adaptive
CRW Simple
GARCH(1,1)
GBM
ACF
Lag



OBI-adaptive coin at OBI=0.75, alpha=0.82
Directional information is phase-encoded; magnitudes remain unitary.







8.7 Multi-horizon distributional distance (BTC)

The scorecard uses h = 1, but the Wasserstein distance is informative across horizons h in {1,10,50,100}. Table 11 reports W[redacted] (×10[redacted] return units); the KS statistic is computed at h = 1 only (longer horizons have too few non-overlapping windows for a reliable two-sample KS, hence reported as not-available).

Table 11 — Wasserstein distance W■ by horizon (BTC, ×10■■). Smaller is better; best per column in bold.

Model	h = 1	h = 10	h = 50	h = 100
QRW Heavy-Tail	**0.60**	0.7	0.027 (27)	0.049 (49)
QRW Adaptive	0.72	0.7	0.036 (36)	0.056 (56)
CRW Simple	0.70	0.7	0.035 (35)	0.056 (56)
CRW Biased	0.70	0.7	0.035 (35)	0.056 (56)
CRW Correlated	0.70	0.7	0.035 (35)	0.056 (56)
GARCH(1,1)	5.60	13	**0.022 (22)**	**0.026 (26)**
GBM	2.85	0.7	0.028 (28)	0.051 (51)

Insight. At the **shortest horizon (h = 1)** the heavy-tailed QRW has the smallest transport distance; at **longer horizons (h = 50, 100)** GARCH overtakes, because its volatility-clustering structure better captures the growth of the return distribution's spread over many steps, while the (memory-less-magnitude) QRW and CRW accumulate variance roughly linearly. Point-forecast superiority of the QRW is thus a *short-horizon* phenomenon — consistent with the microstructure intuition that order-book imbalance predicts the *imminent* move, not the cumulative path far ahead.

8.8 Full Diebold–Mariano matrix (BTC)

Table 12 — All 21 pairwise Diebold–Mariano comparisons (mean-path absolute loss, Newey–West HAC L = 20, BH-adjusted p). "Winner" is the lower-loss model; all comparisons are significant at the BH-adjusted 5% level.

Model 1	Model 2	DM stat	p (BH)	Winner
QRW Adaptive	QRW Heavy-Tail	+3.21	0.0014	QRW Heavy-Tail
QRW Adaptive	CRW Simple	–3.89	0.0003	QRW Adaptive
QRW Adaptive	CRW Biased	–3.90	0.0003	QRW Adaptive
QRW Adaptive	CRW Correlated	–3.89	0.0003	QRW Adaptive

QRW Adaptive	GARCH(1,1)	–3.90	0.0003	QRW Adaptive
QRW Adaptive	GBM	–3.95	0.0001	QRW Adaptive
QRW Heavy-Tail	CRW Simple	–3.33	0.0011	QRW Heavy-Tail
QRW Heavy-Tail	CRW Biased	–3.33	0.0011	QRW Heavy-Tail
QRW Heavy-Tail	CRW Correlated	–3.33	0.0011	QRW Heavy-Tail
QRW Heavy-Tail	GARCH(1,1)	–3.37	0.0011	QRW Heavy-Tail
QRW Heavy-Tail	GBM	–3.31	0.0012	QRW Heavy-Tail
CRW Simple	CRW Biased	–5.22	4×10■■	CRW Simple
CRW Simple	CRW Correlated	–3.89	0.0003	CRW Simple
CRW Simple	GARCH(1,1)	–3.23	0.0014	CRW Simple
CRW Simple	GBM	+3.65	0.0005	GBM
CRW Biased	CRW Correlated	+4.28	0.0002	CRW Correlated
CRW Biased	GARCH(1,1)	–3.18	0.0015	CRW Biased
CRW Biased	GBM	+3.68	0.0005	GBM
CRW Correlated	GARCH(1,1)	–3.20	0.0014	CRW Correlated
CRW Correlated	GBM	+3.66	0.0005	GBM
GARCH(1,1)	GBM	+3.64	0.0005	GBM

Insight. The DM tournament induces a clear **mean-path skill ordering**: QRW Heavy-Tail ■ QRW Adaptive ■ {CRW Simple ≈ CRW Correlated ■ CRW Biased} ■ GBM ■ GARCH. The two QRW models are the only ones that beat *every* classical baseline on mean-path loss, and the heavy-tailed variant beats the base QRW. This is the exact inverse of the distributional scorecard ordering (Table 5), crystallising the paper's central tension: **the quantum-walk coin is the best mean-path/directional predictor while being among the worst distributional fits.**

> **Tóm t■t 8.5–8.8 (VI):** Per-asset: trên **ETH** QRW Heavy-Tail **#1** (kh■p KS & Wasserstein t■t nh■t) vì ETH khu■ch tán (β≈0,95) h■p v■i QRW; trên **BNB** (■uôi n■ng, siêu khu■ch tán) CRW Correlated th■ng, QRW Heavy-Tail #3.

S■a tail-index rõ trên BTC, m■ trên ETH/BNB (■uôi các tài s■n này nh■ h■n). GARCH/GBM luôn b■ bác KS m■nh nh■t ■ thang tick. ■a-horizon: QRW t■t nh■t Wasserstein ■ h=1, GARCH v■■t ■ h=50/100 (■u th■ QRW là *ng■n h■n*). Ma tr■n DM: **QRW Heavy-Tail ■ QRW Adaptive ■ CRW ■ GBM ■ GARCH** v■ d■ báo ■■■ng — ■■o ng■■c th■ t■ th■ ■i■m phân ph■i.

8.9 Which model wins which criterion (BTC)

Aggregating all tests, Table 13 records the *single best* model per evaluation criterion — the clearest statement of the "no single winner" conclusion.

Table 13 — Per-criterion winner (BTC, n_paths = 3000).

Criterion	What it measures	Winner	Runner-up
KS distribution (h=1)	Distributional shape	CRW Correlated	CRW Biased
Wasserstein (h=1)	Transport / location	**QRW Heavy-Tail**	CRW Correlated
Variance scaling β	Diffusion regime	CRW Correlated	GARCH
ACF MSE	Autocorrelation shape	CRW Correlated	GBM
Hill tail index	Power-law tail	GARCH	**QRW Heavy-Tail**
Excess kurtosis	Peakedness/tails	GARCH	CRW Simple
VaR/CVaR 99%	Tail risk level	GBM	GARCH
Aggregate scorecard	Distribution (mean rank)	CRW Correlated	CRW Biased
Directional AIC (within family)	Sign prediction	**QRW Adaptive**	**QRW Heavy-Tail**
Diebold–Mariano	Mean-path accuracy	**QRW Heavy-Tail**	QRW Adaptive
Bootstrap path-MAE rank	Path accuracy	**QRW Heavy-Tail**	QRW Adaptive

Insight. The winners split cleanly into **three "skill territories"**: *distribution/scaling/autocorrelation* → correlated CRW; *tail level/risk* → GARCH/GBM; *direction and point/path accuracy* → the QRW family. Five of eleven criteria fall to a quantum model, six to classical models — a near-tie that depends entirely on which loss the analyst prioritises. This table operationalises the paper's thesis that **forecasting skill is multi-dimensional and no single model is Pareto-dominant.**

8.10 Monte-Carlo sensitivity (BTC, 2000 vs 3000 paths)

Because the heavy-tailed model's kurtosis/tail ranks depend on extreme Pareto draws, the BTC overall scorecard rank shifts with the Monte-Carlo budget. Table 14 contrasts the two runs.

Table 14 — BTC overall scorecard rank by path budget.

Model	**n_paths = 2000**	**n_paths = 3000**	**Δ**
CRW Correlated	1	1	0
CRW Biased	2	2	0
QRW Heavy-Tail	**3**	**6**	**+3 (worse)**
GBM	4	4 (tie)	0
CRW Simple	5	4 (tie)	−1
QRW Adaptive	6	7	+1
GARCH(1,1)	7	3	−4

Insight. The **stable** anchors are the two leading correlated/biased CRWs (rank 1–2 in both runs) and the diffusive QRW Adaptive (near the bottom). The **volatile** entries are QRW Heavy-Tail (3→6) and GARCH (7→3), precisely the two models whose ranking hinges on the *tail* metrics (kurtosis, VaR/CVaR), which are the most Monte-Carlo-sensitive. The distributional-shape metrics (KS, β, ACF) — and therefore the CRW Correlated victory — are invariant to the path budget. We therefore treat **shape/scaling conclusions as robust** and **tail-rank conclusions as budget-conditional**, and recommend large path budgets with reported rank CIs (Section 7.7) for any tail-sensitive claim. Reporting both runs, rather than the more favourable n_paths = 2000 result, is a deliberate anti-cherry-picking choice.

8.11 Consolidated cross-asset comparison

Bringing the per-asset results together, Table 15 places each model's overall rank side by side with the asset's empirical character, making the data-driven nature of the ranking explicit.

Table 15 — Overall rank by asset (n_paths = 2000) against empirical character.

Model	**BTC (heavy, β1.29)**	**ETH (thin, β0.95)**	**BNB (heavy, β1.26)**	**Pattern**
CRW Correlated	1	5	1	Wins heavy/super-diffusive, loses diffusive
CRW Biased	2	3	2	Consistently strong
QRW Heavy-Tail	3	**1**	3	Best on diffusive ETH; competitive elsewhere
QRW Adaptive	6	2	4	Strong only where data are diffusive
CRW Simple	5	3	5	Mid-table
GBM	4	6	6	Weak except BTC tail-risk
GARCH(1,1)	7	7	7	Worst one-step shape on every asset

Insight. The ranking is **not a fixed property of the models** but an interaction between each model's scaling regime and the asset's empirical scaling. The correlated CRW ($\beta \approx 1.3$) and the diffusive QRW ($\beta \approx 1.0$) are mirror images: each wins on the assets whose empirical β it matches (CRW on super-diffusive BTC/BNB; QRW on diffusive ETH) and loses on the others. This is the single most important *generalisable* lesson of the study: **model selection at the tick scale should be conditioned on the asset's measured scaling exponent**, not assumed universal. GARCH is uniformly last on one-step shape across all three assets — a robust negative result for continuous-Gaussian micro-models.

Tóm t■t 8.11 (VI): B■ng h■p nh■t cho th■y th■ h■ng **không c■ ■■nh theo mô hình** mà là t■■ng tác gi■a regime scaling c■a mô hình và β empirical c■a tài s■n: CRW correlated (β≈1,3) th■ng tài s■n siêu khu■ch tán (BTC/BNB), QRW khu■ch tán (β≈1,0) th■ng tài s■n khu■ch tán (ETH) — hai m■t ■■i x■ng. Bài h■c t■ng quát: **ch■n mô hình ■ thang tick nên d■a trên s■ m■ scaling ■o ■■■c c■a tài s■n**, không gi■ ■■nh ph■ quát. GARCH luôn cu■i v■ d■ng m■t-b■■c trên c■ ba tài s■n.

Part V — Discussion / Ph■n V — Bàn lu■n

9. Discussion / Th■o lu■n

9.1 What the quantum walk does and does not buy

Three robust conclusions survive across assets and Monte-Carlo budgets.

The coin helps direction. The adaptive, OBI-driven coin gives the QRW family the **highest directional Bernoulli likelihood** and the **best mean-path / Wasserstein / Diebold–Mariano** scores. Encoding order-book imbalance into the coin is a genuine, statistically significant improvement for *point and directional* forecasting — consistent with the microstructure literature that OBI predicts short-horizon direction [Cont, Stoikov & Talreja 2010; Gould et al. 2013].

Coherence does not survive calibration. The theoretical hallmark of a QRW — ballistic, super-diffusive spreading — **disappears** once the model is calibrated with its decoherence channel and event clock: the fitted QRW has $\beta \approx 1.0$ (ordinary diffusion), whereas the empirical $\beta \approx 1.28$ super-diffusion is matched instead by a *classical* correlated walk. The quantum advantage that motivates the model is calibrated away; what remains is an adaptive directional predictor whose stochastic structure is effectively classical-diffusive.

Heavy tails require an explicit jump mechanism, not quantum structure. Repairing the tail index needed a Pareto shift, not anything quantum; and that fix overshoots kurtosis. Heavy tails are a property of the *magnitude* process, orthogonal to the coin/interference structure.

9.2 Why frame this as a falsification benchmark

The honest reading of Tables 1–8 is that **the quantum-walk analogy, at the tick scale studied, does not dominate a parsimonious classical baseline on distributional fidelity**, even though it wins on directional/point prediction. Rather than over-claim, we follow the project's own methodological stance (Section 4.1, and the theory note's "rejectable hypotheses"): a clearly-stated negative/mixed result, obtained under a rigorous reproducible protocol, is a valid and useful scientific contribution. The artefact's value is as a **reproducible falsification benchmark**: a fixed protocol, corrected statistics, and seven models against which future QRW variants (2-D walks, learned coins, continuous-time QRWs) can be measured without re-litigating the evaluation design.

9.3 Reconciling the conflicting "winners"

The apparent contradiction — QRW best by DM/Wasserstein/AIC-directional, worst by scorecard — is not a bug but a **decomposition of forecasting skill**. Point/location accuracy (where is the price heading) and distributional/tail accuracy (what is the full risk profile) are distinct objectives served by distinct model components. A practitioner who needs a *directional* signal would prefer the adaptive QRW; one who needs a *risk/distribution* model would prefer the correlated CRW or GARCH. The scorecard, by averaging seven distributional metrics, structurally favours distributional fidelity over point accuracy; we report all loss notions so the reader can weight them by use case.

> **Tóm t■t 9 (VI):** Ba k■t lu■n b■n v■ng: (1) **coin giúp d■ báo h■■ng** (AIC h■■ng, Wasserstein, DM t■t nh■t) — mã hoá OBI vào coin có giá tr■ th■t; (2) **tính l■■ng t■ (ballistic/siêu khu■ch tán) b■ calibration tri■t tiêu** (β QRW ≈1,0; siêu khu■ch tán empirical l■i do CRW correlated tái t■o); (3) **■uôi n■ng c■n c■ ch■ nh■y t■■ng minh** (Pareto), không ph■i c■u trúc l■■ng t■, và b■n s■a l■i v■■t ng■■ng kurtosis. Vì v■y trình bày nh■ **benchmark ki■m ch■ng**: k■t qu■ h■n h■p/âm nh■ng nghiêm ng■t và h■u ích. Mâu thu■n "ai th■ng" th■c ra là **phân rã k■ n■ng d■ báo**: ■i■m/h■■ng vs phân ph■i/■uôi là hai m■c tiêu khác nhau.

9.4 Comparison with prior QRW-in-finance work

Most prior QRW-in-finance contributions are either (a) theoretical/pricing-oriented — e.g. the path-integral "quantum finance" of Baaquie [2004], quantum-game portfolio analogies [Piotrowski & S■adkowski 2004], or surveys of quantum *computing* for finance [Orús, Mugel & Lizaso 2019] — or (b) empirical demonstrations on low-frequency series against a single classical baseline, typically without an ex-ante protocol, multiple-testing correction, or a predictive-accuracy significance test. Relative to that body of work, the present study differs on four axes:

1. **Scale and granularity.** We use 31 days of *tick-level* LOB-proxy data across three assets (>113M BTC events), versus the daily/weekly samples common in earlier empirical QRW work.
2. **Baseline strength.** We benchmark against *five* classical alternatives spanning the directional (simple/biased/correlated RW) and continuous-Gaussian (GARCH, GBM) families, not a single GBM.
3. **Inferential rigour.** We add Diebold–Mariano significance with HAC variance, BH false-discovery control, bootstrap rank intervals, and within-family AIC/BIC — turning rank differences into testable statements.
4. **Stance.** We frame the artefact as a *falsification benchmark* and report a mixed/negative verdict, rather than presenting a single favourable comparison. This directly answers the methodological critique that QRW-finance "claims outpace reproducible evidence" (Section 1.2).

The novel modelling contribution — an OBI-driven adaptive coin combined with a maximum-likelihood Pareto heavy-tailed shift, integrated as a scoreable seventh model — is, to our knowledge, not present in the cited literature; but our honest conclusion is that it does not, at this scale, overturn a parsimonious correlated classical walk on distributional fidelity.

9.5 Practical implications

- **For directional signal extraction**, the adaptive QRW coin is attractive: it has the best within-family directional likelihood and the best short-horizon mean-path accuracy (Tables 7, 11, 12). A practitioner seeking a *next-move* signal from order-book imbalance could use the coin as a calibrated, interpretable predictor.
- **For risk/distribution modelling**, the correlated classical walk (shape, scaling) or GARCH (tails at longer horizons) are preferable; the heavy-tailed QRW's kurtosis overshoot makes it unsuitable as an off-the-shelf VaR engine without recalibration of the jump cap.
- **For research infrastructure**, the released pipeline is reusable: any new model implementing the `simulate`/`fit` interface can be dropped into the seven-model benchmark and scored under the identical ex-ante protocol.

9.6 Threats to validity

- **Construct validity.** OBI is proxied by trade-volume imbalance; the coin's directional edge may change with true L2 depth. The "directional log-likelihood" advantage of the QRW is conditional on this proxy.
- **Internal validity.** A single fixed origin per asset-day, and Monte-Carlo sensitivity of tail metrics (Section 8.4), mean rank differences near the middle of the table are within noise; we mitigate with bootstrap CIs and report both path budgets.
- **External validity.** Three large-cap crypto pairs over one month in 2026; generalisation to equities, FX, other regimes, or other periods is untested. The strong asset-dependence we document (ETH vs BTC/BNB) cautions against extrapolating a single asset's verdict.
- **Statistical-conclusion validity.** Many comparisons are made; we apply Benjamini–Hochberg FDR control throughout, but emphasise effect sizes (ΔAIC, rank gaps, W■ ratios) alongside p-values.

9.7 Why GARCH and GBM fail at the tick scale

A striking, consistent finding is that the two continuous-Gaussian models are the **worst one-step distributional fits on every asset** (KS $p < 10$■■■), even though GARCH is the *best tail* model on BTC. The reconciliation is instructive. GARCH and GBM are *calendar-time*, *continuous-return* models: their one-step innovation is Gaussian with a (possibly time-varying) variance, a description appropriate at minute/day scales after aggregation. At the **event/tick scale**, one-step returns are dominated by the discreteness of the tick grid and by a large mass at (or near) a single tick — a distribution that is neither continuous nor Gaussian. KS, which compares the full one-step CDF, therefore rejects them overwhelmingly. Yet GARCH's variance recursion still captures the *clustering* that drives unconditional kurtosis and the *growth of spread over many steps*, which is why it wins kurtosis (BTC) and longer-horizon Wasserstein (Table 11) despite failing the one-step shape. This dissociation — wrong micro-shape, right macro-volatility — is itself a useful diagnostic: it argues that continuous-Gaussian models should be fit and evaluated at the time scale for which their innovation assumption holds, not naively transplanted to raw tick returns. The discrete-walk family (QRW, CRW), built natively on the tick grid, is the appropriate micro-scale primitive; the open question this paper leaves is which *magnitude* process on that grid best matches the data without the kurtosis overshoot of the capped-Pareto shift.

9.8 Implications for the quantum-finance research programme

Our results carry a constructive message for the broader quantum-finance agenda. The recurring critique of QRW-in-finance is methodological, not conceptual: studies tend to (a) compare against a single weak baseline, (b) omit predictive-accuracy significance testing, (c) evaluate in-sample or without a fixed forecast origin, and (d) report a single favourable configuration. Each of these is avoidable. By pairing every quantum interpretation with a falsifiable classical baseline, applying Diebold–Mariano with HAC variance and FDR control, enforcing an ex-ante fixed-origin protocol, and reporting two Monte-Carlo budgets and three assets, we show that a QRW can be evaluated to the same standard as any econometric model — and that, so evaluated, its advantage is *real but localised* (direction and short-horizon point accuracy) rather than global. This reframes the productive research question from "is the market quantum?" (a metaphysical, untestable framing) to "does a quantum-walk-structured predictor add measurable, out-of-sample skill on a specified loss, relative to a parsimonious classical alternative?" (an empirical, falsifiable one). We contend that future QRW-finance papers should report against a benchmark of this kind; the artefact we release is one concrete instantiation, deliberately designed so that a new model is a drop-in addition scored under identical conditions.

> **Tóm t■t 9.8 (VI):** Thông ■i■p xây d■ng cho ch■■ng trình quantum-finance: phê phán lâu nay là v■ *ph■■ng pháp* (baseline y■u, thi■u ki■m ■■nh ý ngh■a d■ báo, ■ánh giá in-sample, ch■ báo c■u hình thu■n l■i) — ■■u kh■c ph■c ■■■c. Khi ■ánh giá ■úng chu■n (baseline c■ ■i■n có th■ bác b■, DM + HAC + FDR, giao th■c ex-ante g■c c■ ■■nh, hai budget, ba tài s■n), l■i th■ QRW là **th■t nh■ng c■c b■** (h■■ng + ■i■m ng■n h■n), không ph■ quát. Câu h■i nên chuy■n t■ "th■ tr■■ng có l■■ng t■ không?" (siêu hình, không ki■m ■■■c) sang "predictor c■u trúc quantum-walk có thêm k■ n■ng out-of-sample ■o ■■■c trên m■t loss xác ■■nh, so v■i baseline c■ ■i■n ti■t ki■m tham s■ không?" (th■c nghi■m, bác b■ ■■■c). Benchmark công b■ là

m■t hi■n th■c hoá c■ th■.

Tóm t■t 9.7 (VI): GARCH/GBM **luôn b■ bác KS m■nh nh■t ■** thang tick vì chúng là mô hình **l■i su■t liên t■c, th■i gian l■ch** (Gaussian) — không h■p v■i l■■i tick r■i r■c có kh■i l■n quanh m■t tick. Nh■ng GARCH v■n b■t **gom c■m bi■n ■■ng** nên th■ng kurtosis (BTC) và Wasserstein dài h■n: *sai hình vi mô, ■úng bi■n ■■ng v■ mô*. Hàm ý: nên fit/■ánh giá mô hình Gaussian ■ thang phù h■p v■i gi■ ■■nh, không áp th■ng lên l■i su■t tick. H■ random-walk r■i r■c (QRW/CRW) m■i là nguyên thu■ ■úng cho vi mô; câu h■i m■: quá trình *magnitude* nào trên l■■i tick kh■p d■ li■u mà không v■■t ng■■ng kurtosis nh■ Pareto-có-cap.

10. Limitations / H■n ch■

1. **OBI is a trade-volume proxy.** The public trade archive lacks L2 depth, so OBI is approximated by causal trade-volume imbalance. True top-of-book imbalance may carry stronger directional content; the coin's predictive edge could be under- or over-stated.
2. **Single forecast origin per asset-day.** The fixed-origin protocol scores one holdout window per benchmark; although robust across three assets, a rolling-origin design with many windows would tighten the rank confidence intervals.
3. **Monte-Carlo sensitivity of tail metrics.** As Section 8.4 shows, kurtosis/tail ranks for the heavy-tailed model depend on the path budget; absolute rank claims should be read with this caveat.
4. **Event clock and one-tick base shift.** The base QRW moves one tick per event; richer multi-tick or lazy (stay-put) walks, and a feature-dependent decoherence η_t fitted to spread/latency, are left to future work.
5. **No transaction-cost or execution layer.** The study is about distributional/forecast fidelity, not a trading strategy; directional edge does not imply profitability after costs.
6. **Tail cap is data-bounded.** Capping jumps at the in-sample maximum prevents explosions but also prevents the model from generating genuinely unprecedented moves, biasing extreme-tail risk downward (note the near-zero simulated VaR/CVaR).

> **Tóm t■t 10 (VI):** H■n ch■: OBI ch■ là proxy kh■i l■■ng (thi■u L2 depth); m■t g■c d■ báo m■i tài s■n-ngày (nên dùng rolling-origin); ch■ s■ ■uôi nh■y s■ ■■■ng Monte-Carlo; mô hình b■■c m■t-tick/event clock còn ■■n gi■n (ch■a có multi-tick, lazy walk, η_t theo spread/latency); ch■a có l■p chi phí giao d■ch (edge h■■ng ≠ l■i nhu■n); cap nh■y theo max in-sample làm thiên l■ch r■i ro ■uôi xu■ng.

Part VI — Conclusion and Recommendations / Ph■n VI — K■t lu■n và ki■n ngh■

11. Conclusion and Recommendations / K■t lu■n và ki■n ngh■

We built an adaptive, ex-ante discrete-time coined QRW for cryptocurrency market microstructure, extended it with a maximum-likelihood-calibrated heavy-tailed (Pareto) shift, and benchmarked it against six alternatives on 31 days of BTC/ETH/BNB tick data under a rigorous, reproducible, multiple-testing-corrected protocol. The heavy-tailed extension **repairs the QRW's catastrophic tail-index pathology** (Hill index 1.6×10■ → 1.45 vs 2.47 empirical) and the **QRW family is the best directional/point predictor** (highest within-family directional AIC; best Wasserstein and Diebold–Mariano). Yet the QRW **does not beat a parsimonious correlated classical walk on aggregate distributional fidelity**: it now overshoots kurtosis, and the empirical super-diffusive variance scaling is reproduced by the classical — not the quantum — model, because calibration suppresses the QRW's ballistic mechanism. The pattern is consistent across three assets, modulo a documented Monte-Carlo sensitivity of tail metrics.

We therefore present the work as a **reproducible falsification benchmark** for QRWs in market microstructure: a fixed protocol, corrected statistics, seven models, and an honest, mixed verdict.

11.1 Future-work roadmap

1. **Native coupling of direction and magnitude (2-D / multi-coin walks).** The cleanest deficiency exposed here is that direction (coin) and magnitude (shift) are modelled separately. A two-dimensional or multi-coin walk, where a higher-dimensional coin encodes both the sign and a discretised magnitude class, could couple them natively and might avoid the kurtosis overshoot of the bolted-on Pareto shift.
2. **Tempered / truncated power-law magnitude.** Replacing the capped Pareto with a *tempered* power law (exponential cut-off) or a truncated stable law could match the empirical tail index *and* kurtosis simultaneously, closing the gap identified in Section 8.2. This targets the magnitude law, which Section 6.6 isolates as the sole driver of the QRW's distributional behaviour.
3. **Learned coin.** A neural or kernel parameterisation of the coin angle, regularised against the directional-likelihood baseline and validated out-of-sample,

could exploit richer feature interactions while preserving interpretability through the within-family AIC comparison.

4. **Feature-dependent decoherence η_t.** Fitting η_t = sigmoid(β■ + β■·spread + β■·latency − β■·depth) and testing whether *partial* coherence (an empirical β strictly between 1 and 2) is recoverable out-of-sample would directly probe whether any genuine quantum-like super-diffusion survives at intermediate horizons.
5. **Full L2 order-book imbalance.** Replacing the trade-volume OBI proxy with true top-K depth imbalance (depth snapshots) would test whether the QRW's directional edge strengthens, and would let the coin condition on resting liquidity rather than realised flow alone.
6. **Rolling-origin, large-budget evaluation.** A rolling-origin design with many forecast windows and a large Monte-Carlo budget, reporting bootstrap rank confidence intervals throughout, would tighten the (currently budget-conditional) tail-metric rankings and convert the single-origin snapshots into distributions of skill.
7. **Cross-domain generalisation.** Extending the benchmark to equities, FX and futures, and across volatility regimes, would test the strong asset-dependence documented here (ETH vs BTC/BNB) and establish whether the "QRW wins where data are diffusive" rule holds more broadly.

Each direction is a falsifiable extension that the released benchmark can score under the identical protocol — the artefact's intended purpose.

> **Tóm t■t 11.1 (VI):** B■y h■■ng phát tri■n: (1) walk 2-D/■a coin ghép h■■ng+magnitude t■ nhiên; (2) power-law tempered/c■t c■t ■■ kh■p ■■ng th■i tail-index và kurtosis (s■a v■■t ng■■ng); (3) coin h■c ■■■c (neural/kernel) có regularize theo baseline; (4) decoherence η_t theo spread/latency ■■ dò coherence *m■t ph■n* (β gi■a 1 và 2); (5) OBI t■ L2 depth th■t; (6) ■ánh giá rolling-origin nhi■u c■a s■ + budget l■n + kho■ng tin c■y h■ng; (7) t■ng quát hoá sang c■ phi■u/FX/phái sinh ■■ ki■m quy lu■t "QRW th■ng n■i d■ li■u khu■ch tán". M■i h■■ng ■■u ch■m ■■■c b■ng benchmark ■ã công b■ theo cùng giao th■c.

Tóm t■t 11 (VI): QRW thích nghi + b■■c ■uôi n■ng Pareto **s■a ■■■c l■i tail-index** và là **b■ d■ báo h■■ng/■i■m t■t nh■t**, nh■ng **không v■■t CRW t■■ng quan ■** phân ph■i t■ng h■p (v■■t ng■■ng kurtosis; siêu khu■ch tán do CRW tái t■o, không ph■i QRW — vì calibration tri■t tiêu c■ ch■ ballistic). Nh■t quán trên 3 tài s■n, kèm l■u ý nh■y Monte-Carlo. Công b■ nh■ **benchmark ki■m ch■ng tái l■p**. H■■ng t■i: QRW 2-D/■a coin, coin h■c ■■■c, CTQW, decoherence η_t theo spread/latency, OBI t■ L2 ■■y ■■, ■ánh giá rolling-origin nhi■u ■■■ng.

Part VII — References and Appendices / Ph■n VII — Tài li■u tham kh■o và Ph■ l■c

12. References / Tài li■u tham kh■o

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Canonical methodological and foundational works

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Note on referencing integrity / L■u ý v■ tính trung th■c c■a trích d■n: No reference here is fabricated. Entries [1]–[22] reproduce the project's DOI-verified literature review; entries [23]–[31] are universally cited standard works whose bibliographic details are well established. Readers should resolve each DOI to confirm.

Appendix A — Artefact Catalogue: meaning of every file, chart and script / Ph■ l■c A — Gi■i ngh■a t■ng t■p, bi■u ■■, script

This appendix documents what each produced artefact *means* and what *insight* it carries, as requested for full reproducibility and interpretability.

A.1 Statistical result files (`results/`)

File	Contents	Meaning / insight (EN)	Ý ngh■a (VI)
`scorecard.csv`	Per-metric ranks + mean/overall rank, 7 models	The headline distributional ranking; CRW Correlated #1, QRW variants last	Th■ ■i■m phân ph■i t■ng h■p; CRW Correlated nh■t, QRW cu■i
`tail_analysis.csv`	Hill tail index, kurtosis, VaR/CVaR 99%	Shows fixed-tick tail pathology (~10■) and the heavy-tail repair (1.45) + kurtosis overshoot	L■i tail c■ ■■nh (~10■) và b■n vá ■uôi n■ng (1,45) kèm v■■t kurtosis
`distribution_tests.csv`	KS stat/p, Wasserstein by horizon	Classical walks not rejected (KS); QRW best Wasserstein	CRW không b■ bác (KS); QRW th■ng Wasserstein
`variance_scaling_results.csv`	β exponent + bootstrap CI per model	Empirical super-diffusion β≈1.28 matched only by CRW Correlated	Siêu khu■ch tán β≈1,28 ch■ CRW Correlated kh■p
`autocorrelation_tests.csv`	ACF MSE, Ljung–Box p (lags 1/5/10, BH)	Mild empirical autocorrelation; CRW overshoots, QRW undershoots	T■ t■■ng quan nh■; CRW v■■t, QRW thi■u
`model_comparison_table.csv` / `model_aic_bic_comparison.csv`	AIC/BIC/LL, parameter count, likelihood family	Raw likelihood table feeding model selection	B■ng likelihood thô cho l■a ch■n mô hình
`model_selection_corrected.csv`	AIC/BIC ranked *within* family (issue #8 fix)	QRW best directional AIC; GARCH best continuous	QRW t■t nh■t AIC h■■ng; GARCH t■t nh■t liên t■c

`diebold_mariano_tests.csv`	Pairwise DM stat/p (HAC, BH)	QRW Heavy-Tail significantly best on mean-path loss	QRW Heavy-Tail t■t nh■t d■ báo ■■■ng (có ý ngh■a)
`scorecard_bootstrap_ci.csv`	95% CI of mean rank (path-MAE proxy)	QRW Heavy-Tail rank 1.0; GARCH/GBM worst	QRW Heavy-Tail h■ng 1,0; GARCH/GBM t■ nh■t
`cross_asset_scorecard.csv`	Average ranks across BTC/ETH/BNB	QRW Heavy-Tail ties for best avg rank (2.33)	QRW Heavy-Tail ■■ng h■ng nh■t (2,33)
`cross_asset_summary.json`	Per-asset scorecards + holdout sizes	ETH: QRW Heavy-Tail #1; BTC/BNB: #3	ETH: QRW Heavy-Tail #1; BTC/BNB: #3
`calibrated_params.json`	QRW Adaptive calibrated coefficients	The fitted coin/decoherence parameters	Tham s■ coin/decoherence ■ã hi■u ch■nh
`heavy_tail_calibration.json`	Heavy-tail params incl. tail index	The Pareto α■_tail and jump scale	α■_tail Pareto và jump scale
`garch_params.json`	GARCH(1,1) ω,α,β, persistence, AIC	Volatility-clustering baseline parameters	Tham s■ baseline gom c■m bi■n ■■ng
`benchmark_results.csv`	Raw per-metric values + std, 7 models	The unranked metric values behind the scorecard	Giá tr■ ch■ s■ thô tr■■c khi x■p h■ng
`final_comparison_table.csv`	Merged comparison table	Convenience join of metrics per model	B■ng g■p ti■n tra c■u

A.2 Figures (`figures/`) — what each chart shows

Figure	What it shows	Insight (EN)	Ý ngh■a (VI)
`scorecard.png`	Heatmap/bar of per-metric ranks	Visual summary of Table 5; CRW Correlated dominates	Tóm t■t B■ng 5; CRW Correlated tr■i
`return_distributions.png`	Empirical vs model return densities	QRW Heavy-Tail mass in tails; GARCH/GBM mis-shaped	QRW Heavy-Tail có kh■i ■uôi; GARCH/GBM l■ch d■ng

`variance_scaling.png`	log Var vs log horizon, slope β	Empirical & CRW Correlated steeper (super-diffusive); QRW flat	Empirical & CRW Correlated d■c h■n; QRW ph■ng
`acf_comparison.png`	Return ACF by lag, models vs empirical	Near-zero ACF; CRW Correlated extra persistence	ACF ~0; CRW Correlated d■ trí nh■
`sample_paths.png`	Example simulated price paths	Heavy-tail paths show occasional large jumps	■■■ng ■uôi n■ng có cú nh■y l■n thi tho■ng
`coin_operator_heatmap.png`	Coin matrix entries vs features/OBI	How OBI bends the coin angle θ_t	OBI u■n góc coin θ_t th■ nào
`prob_evolution.gif`	P(x,t) evolving over steps	Interference fronts of the coherent QRW	M■t sóng giao thoa c■a QRW thu■n
`cross_asset_comparison.png`	Heatmap of model×metric across assets	Cross-asset consistency of the ranking pattern	Tính nh■t quán x■p h■ng qua các tài s■n
`dashboard_screenshot.png`	Static preview of the interactive dashboard	Reproducibility/inspection aid	H■ tr■ ki■m tra/tái l■p

A.3 Pipeline scripts (`scripts/`)

Script	**Role**
`phase2_pipeline.py`	Clean ticks, engineer features, data-quality checkpoint
`phase3_pipeline.py`	QRW calibration (structural + selection) and overfitting audits
`phase4_pipeline.py`	Benchmark simulation and metric computation
`phase5_pipeline.py`	Statistical tests + scorecard + within-family model selection
`phase6_pipeline.py`	Figures + final report/slides PDF rendering and verification
`phase7a_data_expansion.py`	Multi-day BTC download/process/featurize (113M-event store)

`phase7b_cross_asset.py`	Full pipeline for BTC/ETH/BNB + cross-asset scorecard
`phase_c_model_improvement.py`	Heavy-tail jump grid search / calibration

A.4 Core source modules (`src/`)

Module	Role
`models/qrw_core.py`	Pure DTQRW / density-matrix engine and invariants
`models/adaptive_market_qrw.py`	OBI-adaptive coin + decoherence, two-stage calibration
`models/qrw_heavy_tail.py`	Heavy-tailed (Pareto) shift extension
`models/classical_rw.py`	Simple/biased/correlated classical walks
`models/garch_model.py`, `models/gbm_model.py`	GARCH(1,1) and GBM baselines
`evaluation/benchmark_suite.py`	Fixed-origin ex-ante 7-model benchmark
`evaluation/statistical_tests.py`	KS/Wasserstein/scaling/ACF/tail/DM/bootstrap/selection
`evaluation/results_compiler.py`	Scorecard assembly and ranking
`reporting/report_builder.py`	Matplotlib PDF report renderer

A.5 Pipeline phase-by-phase (inputs → outputs)

- **Phase 2 — `phase2_pipeline.py`.** *In:* raw daily trade archives. *Out:* cleaned `data/processed/*.parquet`, feature `data/features/*.parquet`, `reports/data_quality_*.txt`, `reports/feature_metadata_*.json`. *Role:* the data foundation; segments at gaps, removes outliers, computes causal features.
- **Phase 3 — `phase3_pipeline.py`.** *In:* features. *Out:* `results/calibrated_params.json`, overfitting-audit reports. *Role:* QRW structural+selection calibration with explicit low-sample diagnostics.
- **Phase 4 — `phase4_pipeline.py`.** *In:* features. *Out:* `results/benchmark_results.csv`, `reports/checkpoints/phase4_*`. *Role:* simulate all models and compute raw metrics.
- **Phase 5 — `phase5_pipeline.py`.** *In:* features. *Out:* `results/{distribution,variance_scaling,autocorrelation,tail}_tests.csv`,

`scorecard.csv`, `model_selection_corrected.csv`, `diebold_mariano_tests.csv`, `scorecard_bootstrap_ci.csv`. *Role:* the full statistical battery + scorecard (this paper's core numbers).

- **Phase 6 — `phase6_pipeline.py`.** *In:* Phase-5 artefacts. *Out:* `figures/*.png`, `docs/final_report.pdf`, `docs/presentation_slides.pdf`. *Role:* visualisation and report rendering with page-count verification.
- **Phase 7A — `phase7a_data_expansion.py`.** *Out:* 31-day BTC feature store (113M events). *Role:* multi-day expansion for robust calibration.
- **Phase 7B — `phase7b_cross_asset.py`.** *Out:* `results/{ethusdt,bnbusdt,btcusdt}/*`, `cross_asset_scorecard.csv`, `cross_asset_summary.json`. *Role:* full cross-asset robustness benchmark (7 models × 3 assets).

A.6 Data files (`data/`, git-ignored)

Path	Contents
`data/raw/tick_*.csv.gz`	Canonical gzip trade archives per day
`data/processed/tick_processed_*.parquet`	Cleaned, segmented ticks
`data/features/features_*.parquet`	Engineered features (one file per asset-day)

These are reconstructed deterministically by the Phase-2/7A scripts and are excluded from version control because of size (the BTC feature store alone exceeds 100M rows).

Appendix B — Extended tables / Ph■ l■c B — B■ng m■ r■ng

B.1 Calibrated QRW Adaptive parameters (BTC)

Parameter	Value	Interpretatio
γ (decoherence base)	0.143	Baseline dephasing strength
ρ■ (persistence)	0.867	Directional memory of the coin
obi_bias	1.175	Intercept of OBI response
α_OBI	0.437	Coin sensitivity to OBI
α_direction	–1.147	Sensitivity to signed tick direction
α_ΔOBI	–0.629	Sensitivity to OBI change
α_	OBI	
γ_intensity	–0.951	Decoherence response to trade intensity

p_move	0.173		Unconditiona one-step move probability

B.2 GARCH(1,1) calibrated parameters (BTC)

Parameter	Value
ω	2.08×10■^{12}
α (ARCH)	0.674
β (GARCH)	0.313
persistence α+β	0.987
log-likelihood	1.330×10■
AIC / BIC	–2.660×10■ / –2.658×10■
observations	1143 (scaled returns ×10■)

B.3 Per-asset holdout sizes (cross-asset study)

Asset	Holdout events	Top model	QRW Heavy-Tail rank
BTC/USDT	1,246,144	CRW Correlated	3
ETH/USDT	1,149,168	QRW Heavy-Tail	1
BNB/USDT	224,803	CRW Correlated	3

Appendix C — Reproducibility / Ph■ l■c C — Kh■ n■ng tái l■p

Environment. Python 3.13; numpy, pandas, scipy, matplotlib, pyarrow, h5py, loguru, PyYAML (see `requirements.txt`). All randomness is seeded (`random_seed = 2026`).

Data. `python scripts/phase7a_data_expansion.py` (BTC) and `python scripts/phase7b_cross_asset.py` (ETH/BNB + cross-asset) download from `https://data.binance.vision` and build the feature stores. Raw/processed/feature data are git-ignored (large); re-running the scripts reconstructs them deterministically.

Single-asset benchmark (this paper's BTC deep run).

```
python scripts/phase5_pipeline.py \ --feature-path
data/features/features_BTCUSDT_2026-06-12.parquet \ --n-paths 3000 --n-steps 500
--bootstrap-iterations 500 python scripts/phase6_pipeline.py \ --feature-path
data/features/features_BTCUSDT_2026-06-12.parquet \ --n-paths 3000 --n-steps 500
```

Cross-asset robustness. `python scripts/phase7b_cross_asset.py --n-paths 2000 --n-steps 500`.

Tests. `python -m pytest` (70 tests pass), including the ex-ante leakage test and the heavy-tail benchmark test.

Protocol version. `fixed_origin_ex_ante_zero_inflated_v2`. **Significance.** α = 0.05 with Benjamini–Hochberg FDR correction.

> **Tóm t■t ph■ l■c (VI):** Ph■ l■c A gi■i ngh■a t■ng t■p k■t qu■, t■ng bi■u ■■, t■ng script và module — kèm insight. Ph■ l■c B li■t kê tham s■ hi■u ch■nh QRW/GARCH và kích th■■c holdout t■ng tài s■n. Ph■ l■c C ghi môi tr■■ng, l■nh tái l■p (seed 2026), giao th■c `fixed_origin_ex_ante_zero_inflated_v2`, và 70 test pass. Toàn b■ con s■ trong bài ■■u sinh t■ pipeline này.

Appendix D — Glossary / Ph■ l■c D — Thu■t ng■

Term	Definition (EN)	■■nh ngh■a (VI)
Amplitude	Complex number whose squared modulus gives a probability	S■ ph■c, bình ph■■ng mô-■un cho xác su■t
Ballistic spreading	σ grows ~ t (Var ~ t^2)	Lan truy■n v■i σ ~ t (Var ~ t^2)
Diffusive spreading	σ grows ~ √t (Var ~ t)	Lan truy■n v■i σ ~ √t (Var ~ t)
Coin operator	Unitary acting on the internal 2-state space	Toán t■ unitary trên không gian 2 tr■ng thái n■i
Decoherence / dephasing	CPTP channel erasing off-diagonal coherence	Kênh CPTP xoá coherence ngoài ■■■ng chéo
OBI	Order-book imbalance in [−1, 1]	M■t cân b■ng s■ l■nh in [−1, 1]
Hill tail index	Power-law tail exponent estimate	■■c l■■ng s■ m■ ■uôi lu■ th■a
Variance-scaling β	Exponent in Var ~ t■	S■ m■ trong Var ~ t■
KS statistic	Max CDF distance between two samples	Kho■ng cách CDF l■n nh■t gi■a hai m■u
Wasserstein distance	Optimal-transport distance between distributions	Kho■ng cách v■n chuy■n t■i ■u gi■a phân ph■i
Diebold–Mariano test	Test of equal predictive accuracy	Ki■m ■■nh ■■ chính xác d■ báo b■ng nhau
HAC variance	Heteroskedasticity & autocorrelation consistent variance	Ph■■ng sai v■ng v■i ph■■ng sai thay ■■i & t■ t■■ng quan
FDR / Benjamini–Hochberg	False-discovery-rate multiple-testing control	Ki■m soát t■ l■ phát hi■n sai khi ■a ki■m ■■nh
Ex-ante protocol	Forecast uses only pre-origin information	D■ báo ch■ dùng thông tin tr■■c g■c
Falsification benchmark	Reproducible test designed to *reject* claims	Benchmark tái l■p nh■m *bác b■* tuyên b■

Appendix E — Declarations / Ph■ l■c E — Tuyên b■

Data availability. Raw tick archives are publicly available from Binance Public Data (`https://data.binance.vision/data/spot/daily/trades`). All derived artefacts (calibrated parameters, statistical-test outputs, figures) are regenerated deterministically by the released pipeline under fixed seeds; see Appendix C.

Code availability. The full pipeline (`scripts/phase2–phase7`, `src/`), the seven-model benchmark, the statistical-test suite and the PDF renderer (`scripts/render_paper_pdf.py`) accompany this manuscript.

Reproducibility statement. Every numerical value, table and figure in this paper is produced by `python scripts/phase5_pipeline.py` / `phase6_pipeline.py` / `phase7b_cross_asset.py` with `random_seed = 2026`; the test suite (`python -m pytest`, 70 tests) passes, including the ex-ante leakage test and the heavy-tail benchmark test.

Referencing integrity. No reference is fabricated; entries [1]–[22] reproduce the project's DOI-verified literature review and [23]–[31] are universally cited standard works. Readers are invited to resolve each DOI.

Author contributions. [To be completed — e.g., conceptualisation, methodology, software, formal analysis, writing.]

Funding. [To be completed / Không có / None to declare.]

Conflicts of interest. The authors declare no competing financial or non-financial interests.

Ethics. The study uses only publicly available, aggregated market trade data; no human-subjects or personal data are involved.

> **Tóm t■t ph■ l■c E (VI):** D■ li■u thô công khai (Binance); m■i artefact tái sinh xác ■■nh b■ng pipeline (seed 2026); 70 test pass. **Không b■a reference** — [1]–[22] t■ literature review ■ã verify DOI, [23]–[31] là công trình kinh ■i■n. Các m■c ■óng góp tác gi■/tài tr■ ■■ tr■ng ch■ ■i■n; không xung ■■t l■i ích; ch■ dùng d■ li■u th■ tr■■ng công khai.

Appendix F — Protocol checklist and metric quick-reference / Ph■ l■c F — Danh m■c giao th■c và tra c■u ch■ s■

F.1 Ex-ante protocol checklist

A reviewer can verify the evaluation is leakage-free against this checklist:

- [x] Chronological split; calibration uses only the first 60% of each record.
- [x] Forecast origin fixed at the train/holdout boundary; holdout is the *next* 500 moving events.
- [x] No holdout covariate enters any forecast (asserted by `test_qrw_forecast_does_not_use_holdout_covariates`).
- [x] Features are strictly causal (rolling windows end at t; no forward fill from the future).
- [x] All models share seeds, origin, horizon, scoring (`random_seed = 2026`).
- [x] Returns are never differenced across a `segment_id` gap.
- [x] Multiple-testing corrected (Benjamini–Hochberg) within each test family.
- [x] Both Monte-Carlo budgets (2000, 3000) reported; tail-sensitive ranks flagged as budget-conditional.
- [x] Protocol version recorded in diagnostics: `fixed_origin_ex_ante_zero_inflated_v2`.

F.2 Metric quick-reference

Metric	**Family**	**Direction**	**Empirical target**	**Best model (BTC)**
KS statistic	Distribution shape	smaller	$p > 0.05$	CRW Correlated
Wasserstein	Transport	smaller	0	QRW Heavy-Tail
Variance-scaling β	Scaling regime	match	1.285	CRW Correlated
ACF MSE	Autocorrelation	smaller	0	CRW Correlated
Hill tail index	Tail	match	2.47	GARCH

Excess kurtosis	Tail/peak	match	74.8	GARCH
VaR/CVaR 99%	Tail risk	match	1.03/1.76×10■■	GBM
Directional AIC	Sign likelihood	smaller (in family)	—	QRW Adaptive
Diebold–Mariano	Mean-path	lower loss	—	QRW Heavy-Tail

F.3 One-command reproduction

```
# 1. Build the multi-day stores (downloads from Binance; ~hours) python
scripts/phase7a_data_expansion.py python scripts/phase7b_cross_asset.py --n-paths
2000 --n-steps 500 # 2. Single-asset BTC deep run (this paper's headline tables)
python scripts/phase5_pipeline.py --feature-path
data/features/features_BTCUSDT_2026-06-12.parquet \ --n-paths 3000 --n-steps 500
--bootstrap-iterations 500 python scripts/phase6_pipeline.py --feature-path
data/features/features_BTCUSDT_2026-06-12.parquet \ --n-paths 3000 --n-steps 500
# 3. Render this paper to PDF python scripts/render_paper_pdf.py # 4. Run the
test suite (70 tests) python -m pytest
```

> **Tóm t■t ph■ l■c F (VI):** F.1 — danh m■c ki■m tra giao th■c ex-ante (split th■i gian, g■c c■ ■■nh, không rò r■ holdout, ■■c tr■ng nhân qu■, chung seed, không tính l■i su■t xuyên gap, hi■u ch■nh BH, báo c■ 2 budget Monte-Carlo). F.2 — b■ng tra c■u nhanh: m■i ch■ s■ thu■c h■ nào, h■■ng t■t, m■c tiêu empirical, model th■ng (BTC). F.3 — chu■i l■nh tái l■p m■t phát (phase7a/7b → phase5/6 → render → pytest).

End of manuscript / H■t.